MIL-S-16036D(SHIPS)
17 November 1958
SUPERSEDING
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4 January 1957

MILITARY SPECIFICATION

SWITCHBOARDS, POWER, NAVAL SHIPBOARD POWER AND LIGHTING CIRCUIT

1. SCOPE

1.1 Scope.- This specification covers the requirements for electric power and lighting switchgear groups and switchboards for shipboard use.

1.2 Classification.- Electric power and lighting switchboards shall be furnished in the following types as specified (see 6.1):

Type I - Dead-front, deck mounted.
Type II - Live-front, deck mounted.
Type III - Miscellaneous.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-A-270 - Aluminum Alloy, Bars, Rods, and Shapes, Extruded; 6061 and 6062.
QQ-A-318 - Aluminum Alloy 52S - Plate and Sheet.
QQ-A-325 - Aluminum Alloy 61S, Bars, Rods, Shapes and Wire, Rolled or Drawn.
QQ-A-359 - Aluminum Alloy 3S; Plate and Sheet.
QQ-C-504 - Copper: Bus Bars (Including Bars, Rods, and Shapes).
QQ-S-365 - Silver Plating (Electrodeposited).
QQ-S-571 - Solder: Lead alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.
QQ-S-692 - Steel Sheets, Carbon, Cold-Rolled.
QQ-S-693 - Steel, Sheet, Hot-Rolled.
TT-P-664 - Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-R-26 - Resistors, Fixed, Wirewound, (Power Type).
MIL-P-78 - Plastic-Material, Laminated, Thermosetting (for Designation Plates).
MIL-C-490 - Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin-Composition, Nonrigid.
JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-B-857 - Bolts, Nuts, and Studs.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for

FSC 6110

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MILITARY (cont'd.)

MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings; Production, Electrical and Mechanical Equipment Naval Shipboard.
MIL-I-1361 - Instruments, Electrical-Measuring: Shunts, Resistors, and Transformers (Ship and Shore Use).
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-C-2194 - Cables, Electric (Reduced Diameter Types) (Shipboard Use).
MIL-R-2729 - Regulator Sets, Voltage, A.C. Generator Naval Shipboard Use.
MIL-T-2867 - Thermometers, Indicating Resistance.
MIL-I-3158 - Insulation Tape, Electrical Glass-Fiber (Resin-Filled); and Cord, Fibrous-Glass.
MIL-F-5591 - Fasteners; Panel.
MIL-I-7798 - Insulation Tape, Electrical, Pressure-Sensitive Adhesive Plastic.
MIL-P-8585 - Primer; Corrosion-Inhibiting, Low Moisture Sensitivity for Aircraft Use.
MIL-P-15024 - Plates Identification — Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-P-15037 - Plastic-Material, Laminated, Thermosetting, Sheets, Glass-Cloth, Melamine-Resin.
MIL-M-15071 - Manual, Technical for Mechanical and Electrical Equipment (Less Electronics).
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-T-15108 - Transformers, Power, Step-Down, Single Phase, 60-Cycle, 1-KVA Approximate Minimum Rating; and Reactors (Balance Coils)-Dry, Naval Shipboard.
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-F-15160 - Fuses; Instrument, Power and Telephone.
MIL-S-15291 - Switches, Rotary, Snap Action, Bureau Design.
MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).
MIL-P-16018 - Packaging of Switchboards, Controllers, and Motor Starters (Over 40 Pounds).
MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-M-16125 - Meters, Electrical Frequency.
MIL-M-16173 - Corrosion Preventive, Solvent Cutback, Cold-Application.
MIL-T-16315 - Transformers, Power, Step-Down (Miscellaneous, Naval Shipboard Use).
MIL-E-16366 - Electrical Clamps, Lug Terminals and Conductor Splices Pressure Grip.
MIL-F-17292 - Format for List of Repair (Spare) Parts for Shipboard Mechanical and Electrical Equipment.
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Enclosure (Shipboard Use).
MIL-I-17384 - Insulating Compound, Electrical, Quick-Drying.
MIL-C-17587 - Circuit Breaker, Air, Electric, Open Frame (Shipboard Use).
MIL-C-17588 - Circuit Breaker, (Automatic-ALB) and Switch, Toggle (Circuit Breaker, Non-Automatic-NLB), Air, Insulated Enclosure, 125 Volts and Below, AC or DC, Naval Shipboard.
MIL-P-17721 - Plastic Sheet, Laminated, Thermosetting, Glass Mat, Melamine Resin.
MIL-S-17773 - Switching Unit, Power Transfer (Bus Transfer) (Naval Shipboard Use).
MIL-M-17999 - Metal, Expanded, Aluminum.
MIL-S-18396 - Switches, Rotary, Meter and Control, Naval Shipboard.
MIL-W-19088 - Wattmeters, Switchboard Type, 4-1/2 Inch.
MIL-B-19780 - Bus Bar, Aluminum Alloy.

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NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the U.S. Navy.

DRAWINGS

BUREAU OF SHIPS

9-S-4907-L - Type Scale Markings for SWBD Instruments.
9-S-5288-L - Terminal Block, SP for Casualty Power Supply System (DC), 250-V.
9-S-5302-L - Riser Terminal, Types AM, BM, and BMB for Casualty Power Supply System, 200-AMP, 550-V, 3PH, AC.
9-S-5312-L - Fuse Retainer, Type B, for Ferrule Type Fuses.
9-S-5370-L - Switchboard, Battery Charging and Discharging, for Class 1 and 2 Vessels, Type A, B, and C Units, General Arrangement, 125-V, DC.
9000-S6201-74453 - Navy Standard Switchboard Units.
9000-S6202-73063 - Lock and Fuse Retainers for Commercial Knife Switches Type Plan.
9000-S6202-73907 - SWBD Indicator Light - 2-Lamp - SPF.
9000-S6202-74228 - Fuse Holder, Double, Dead Front-Blown Fuse Indicating for 1-1/4 x 1/4 Inch Fuses, Type 10FH2, 30 Amp. 120 Volts.
9000-S6202-74229 - Fuse Holder, Single, Dead Front Blown Fuse Indicating for 1-1/4 x 1/4 Inch Fuses, Type 10FH1, 120 Volts.
9000-S6202-74230 - Fuse Holder, Single, Dead Front-Blown Fuse Indicating for 1-1/2 x 13/32 Inch Fuses, Type 12FH1, 450 Volts.
9000-S6202-74307 - Switchboard Indicator Light 2 Lamp for D.C. Circuit Type B 39A.
9000-S6205-74338 - Navy Standard Type ACB 600 Amp. and 1600 Amp. Circuit Breaker, Draw-Out Mechanism.
9000-S6401-74457 - Lighting Fixture, 32 Watt, SPF, Types JF50A and JF50B.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Definitions of terms. - The following definitions shall apply to the terms listed below wherever such terms appear in this specification:

3.1.1 Standardized unit. - A standardized switchgear unit is a vertical subdivision of a section complete with separate front panels and all required appurtenances constructed in accordance with Drawing 9000-S6201-74453.

3.1.2 Section. - A switchgear section is a self-contained assembly of switchgear units consisting of a complete framework with all required enclosures and with a common base mounting.

3.1.3 Switchgear group. - A switchgear group is synonymous with the term switchboard; however, switchboard groups may consist of two or more individual sections rather than all units being assembled together into a single structure.

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3.1.4 Control benchboard. - A control benchboard is a switchgear section consisting of a switchboard and a control desk combined in a single structure with the top of the control desk, either horizontal or inclined, attached to the front switchboard panels. It is provided with the necessary instruments and control and indicating apparatus for the centralized control of generator, bus-tie, and feeder circuits (also to include the battery circuits on submarines) of one or more switchgear groups.

3.1.5 Load center. - A load center distribution section is a section consisting of one or more distribution units. It serves as a power distribution center in locations requiring power supplies exceeding the capacity of distribution panels.

3.1.6 Nonstandard switchgear. - Nonstandard switchgear is switchgear consisting of units, sections, and groups, not conforming to the standardization requirements of Drawing 9000-S6201-74453.

3.2 General. - The switchboards shall be in accordance with the following general requirements of Specification MIL-E-917. When a requirement of Specification MIL-E-917 conflicts with a requirement of this specification, the requirement of this specification shall govern:

Materials
Metals or coating
Electrical insulation
Soldering
Brazing
Welding
Threaded parts
Inclination

3.2.1 Whether or not specifically mentioned for the various units, all switchgear groups shall be furnished complete with all appurtenances required to accomplish the function specified hereinafter.

3.3 Ship's service power switchgear group for vessels having alternating current (a. c.) power. -

3.3.1 A.c. generator control unit for switchgear groups having a control benchboard. - The following appurtenances shall be mounted on or within the unit:

(a) One generator circuit breaker, type ACB, 500 V., 3-pole, electrically operated, drawout mounted. Electrically operated circuit breakers shall be provided with a minimum of 21 secondary contacts, 6 of these to be reserved for the closing circuit.
(b) Necessary disconnecting devices for disconnecting cables between switchgear sections, and performing other switching functions.
(c) One control power transformer 450/120 V., in accordance with Specification MIL-T-15108 or MIL-T-16315 as applicable, connected to generator side of circuit breaker, for generator circuit breaker and governor motor control circuits.
(d) Potential transformers, for metering, relaying, and synchronizing, in number as required.
(e) Current transformers, for metering and relaying, in number as required.
(f) One current transformer, complete with protective device for remote metering, where remote metering is required.
(g) Fuse panel for fusing of the potential and control transformer primary circuits and any control circuits extending outside of the section.
(h) Space, mounting and wiring, for the voltage regulator equipment, including the current and potential transformers as specified in Specification MIL-R-2729 and for the exciter field rheostat conforming to Specification MIL-R-15109. All equipment shall be furnished with the generators.

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(i) One circuit breaker, type AQB or AQB-LF, 500 V., 3-pole, 250-ampere frame size, with element rated at 250-amperes. Instantaneous trip settings shall be as follows: AQB circuit breaker - 3400 amperes, AQB-LF circuit breaker 3000 amperes. For generators rated 1000 kw. and below, an AQB circuit breaker shall be furnished unless otherwise specified in the contract or order. For generators rated above 1000 kw., an AQB-LF fused circuit breaker shall be furnished. The circuit breaker shall be connected to the generator bus ahead of the generator circuit breaker.

(j) One casualty power terminal, riser type, conforming to Drawing 9-S-5302-L connected through the AQB or AQB-LF circuit breaker, item (i).

(k) One reverse power relay, single phase, to prevent the generator from motoring on the bus after loss of driving power from prime mover.

(l) One indicator light (blue) to show the closed position of the generator circuit breaker.

(m) One generator running and up to voltage relay.

3.3.1.1 When the generator capacity is in excess of 400 kw., 2 each of 3.3.1 (i) and (j) shall be furnished.

3.3.2 A.c. generator control unit for switchgear groups not having a control benchboard. - Appurtenances listed in 3.3.1 shall be mounted on or within the unit. Item (a) shall be manually operated for AQB-A250, AQB-A400 and 640 ampere frame size type ACB generator circuit breakers. Type AQB-A250 and AQB-A400 circuit breakers are not drawout mounted but are removable.

3.3.2.1 When the generator capacity is in excess of 400 kw., 2 each of 3.3.1 (i) and (j) shall be furnished.

3.3.2.2 The following devices shall be mounted on the front panel of the generator unit:

(a) One a.c. ammeter, for reading one phase of generator current.

(b) One a.c. voltmeter with transfer switch, for reading one phase of generator potential and one phase of each bus tie voltage.

(c) One polyphase wattmeter, for reading generator output.

(d) One frequency meter for reading the frequency of the generator.

(e) One synchroscope and one set of synchronizing lights with transfer switch for synchronizing the generator with the main bus, and the main bus with the bus tie circuit.

(f) One control switch for control of the generator circuit breaker where an electrically operated breaker is used.

(g) One local control switch for control of the speed adjusting motor on the governor of the generator prime mover.

(h) One transfer switch to transfer the governor motor control from the local control switch to the remote control switch on the a.c. bus tie unit (or control unit) of the control switchgear group. It shall also control the lamp (see 3.3.4.1 (c)), on bus tie unit or control unit to indicate when the remote switch is in control.

(i) One control switch with white indicator light for control of the generator space heaters when provided with the generator.

(j) Control switches, rheostats, and similar equipment that is designed for panel mounting and is required in conjunction with the voltage regulator equipment mounted in the generator unit shall be mounted and operated from the front panel.

(k) For installations in which a bus tie unit is not provided, one set of three indicator lights with a control switch (see 3.3.4.1 (l)) shall be mounted on the front panel of the generator unit.

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3.3.3 A.c. bus tie unit for switchgear groups having a control benchboard. - The following appurtenances shall be mounted within the unit:

(a) One circuit breaker, type ACB, 500 V., 3-pole, electrically operated, drawout mounted. Electrically operated circuit breakers shall be provided with a minimum of 21 secondary contacts, 6 of these to be reserved for the closing circuit.
(b) One feeder circuit breaker, type ACB, 500-volt, 3-pole, drawout mounted.
(c) Necessary disconnecting devices for performing switching functions as may be required.
(d) Potential transformers, for metering, relaying, and synchronizing, in number as required for the bus connections of the switchgear group.
(e) One current transformer for metering the current interchange through the bus tie circuit breaker.
(f) Fuse panel for fusing the potential and control power transformer primary circuits.
(g) Indicator light (blue) to show the closed position of the bus tie and feeder circuit breaker.

One bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus-tie and shore power unit". In addition this unit shall substitute item (h) for the feeder circuit breaker, and add item (i) shown below:

(h) [1]One shore power connection circuit breaker, type ACB, 500 V., 3-pole, drawout mounted. Circuit breakers 1600 H amp. frame size and larger shall be electrically operated and circuit breakers 1600 amp. frame size and smaller shall be manually operated. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. The current transformer (see 3.3.3 (e)), shall be located so that it is in the circuit when feeding power to the local switchgear group. Where a 3,000 AMP frame size circuit breaker is required, a separate unit shall be provided. Manually operated circuit breakers shall be provided with a minimum of 7 secondary contacts. Electrically operated circuit breakers shall be provided with a minimum of 21 secondary contacts, 6 of these to be reserved for the closing circuit.
(i) One indicator light (blue) to show the closed position of the shore power circuit breaker.

[1]To be furnished on only one bus tie unit of any switchgear group.

3.3.4 A.c. bus tie unit for switchgear groups not having a control benchboard. - The following appurtenances shall be mounted within the unit:

(a) One circuit breaker, type ACB, AQB-A250 or AQB-A400; 500 V., 3-pole. Type ACB circuit breakers are drawout mounted and type AQB circuit breakers are not drawout mounted but are removable. Circuit breakers 1600 amp. frame size and larger shall be electrically operated and circuit breakers 640 amp. frame size and smaller shall be manually operated. Manually operated type ACB circuit breakers shall be provided with a minimum of 7 secondary contacts. Electrically operated circuit breakers shall be provided with a minimum of 21 secondary contacts, 6 of these to be reserved for the closing circuit.
(b) One feeder circuit breaker, type ACB or type AQB-A250, 500 volt, 3-pole, drawout mounted. Type AQB-A250 is not drawout mounted.
(c) Necessary disconnecting devices for performing switching functions as may be required.
(d) Potential transformers, for metering, relaying and synchronizing, in number as required for the bus connections of the switchgear group.

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(e) Fuse panel for fusing the control circuits fed from the control power bus.
(f) Fuse panel for fusing the potential and control power transformer primary circuits.

3.3.4.1 The following devices shall be mounted on the front panel of the bus tie unit:

(a) One single element wattmeter, calibrated to indicate the three phase power output of each remote ship's service generator.
(b) One a.c. ammeter for indicating the current of each remote ship's service generator.
(c) Two remote control switches and indicator lights (white) to control the speed adjusting motors on the governors of the prime movers of the remote generators. The control switch power shall be connected through the transfer switches (see 3.3.2.2 (h)) on the generator units of the remote switchgear groups, and the indicating lamps shall be connected to indicate when the transfer of control is accomplished.
(d) One composite transfer switch for reading the power and current of each remote generator by means of (a) and (b) above.
(e) One control switch for control of the bus tie circuit breaker where an electrically operated circuit breaker is used.
(f) Indicator lights (blue) to show the closed position of the bus tie circuit breaker and the type ACB feeder circuit breaker.
(g) One frequency meter for reading the frequency of the bus tie circuit.
(h) One transfer switch, with two white indicator lights, for connecting the switchboard control bus to the local 450 V. ship's service bus, or the 450 V. bus of the corresponding (forward or aft) emergency switchboard. Transfer may be accomplished automatically; however, in such cases one source of power shall be the local 450 V. ship's service bus and the second source shall be a 450-volt bus tie supply.
(i) One set of three indicator lights with a control switch to indicate ground conditions on the 450 volt main bus or any of its connected feeders.

3.3.4.2 One bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus tie and shore power unit". In addition to the equipment specified in 3.3.4 and 3.3.4.1 this unit shall substitute item (a) for the feeder circuit breaker, and add items (b) and (c).

(a) One shore power connection circuit breaker, type ACB, or type AQB-A250, 500 V., 3-pole, drawout mounted. Type AQB-A250 is not drawout mounted. Circuit breakers 1600 H amp. frame size and larger shall be electrically operated and circuit breakers 1600 amp. frame size and smaller shall be manually operated. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. Manually operated type ACB circuit breakers shall be provided with a minimum of 7 secondary contacts. Electrically operated circuit breakers shall be provided with a minimum of 21 secondary contacts, 6 of these to be reserved for the closing circuit.
(b) One indicator light (blue) to show the closed position of the shore power circuit breaker.
(c) Two indicator lights (white), one to indicate the shore power feeder energized and the other to indicate the bus tie feeder energized. These two indicator lights shall be mounted near the shore power circuit breaker together with a warning plate stating: "Do not close this circuit breaker when the bus tie circuit is energized."

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3.3.5 Control benchboard.- This section shall be of the bench type with vertical panels for the mounting of instruments and control equipment, a horizontal or inclined desk top and an apron section below. Control benchboards for surface ships shall be provided with a chair. It shall be so mounted that it can be swung out of the way when not in use. Control switches for the electrically operated circuit breakers controlled from the benchboard shall be mounted on the desk top and arranged within convenient reach of the operator. The manual control rheostats shall be located in the apron portion so as to provide a maximum of free panel space for mounting of equipment. The portion of the apron panel used for mounting of the equipment shall not project beyond the front edge of the desk top. Space shall be provided for filing the framed data sheets.

3.3.5.1 Surface ship control benchboard.- The following devices shall be mounted on the control benchboard for surface ships:

(a) Control switches, rheostats and similar equipment that is designed for panel mounting and is required in conjunction with the voltage regulator equipment mounted in the generator unit, shall be mounted on and operated from the control benchboard.
(b) One a.c. ammeter for each local generator for reading one phase of generator current.
(c) One a.c. voltmeter for each local generator for reading one phase of generator potential.
(d) One a.c. voltmeter with transfer switch for reading single phase voltage of the switchboard bus and of the bus tie feeders.
(e) One polyphase wattmeter for reading generator output for each local generator.
(f) One single element wattmeter with transfer switch calibrated to indicate the 3 phase power output of the remote ship's service generator. (Shall be provided where a switchgear group is to have control of a remote generator or generators.)
(g) One temperature indicator complete with selector switch and accessories provided with the generator.
(h) [1]One frequency meter for reading the frequency of the generator for each local generator.
(i) [1]One frequency meter with transfer switch for reading the frequency of the bus and of each bus tie circuit.
(j) One synchroscope and one set of synchronizing lights with transfer switch for synchronizing the generator with the bus and the bus with the bus tie circuits. Where the switchgear group has control of two generators and two bus tie circuits the synchroscope and synchronizing lights shall be connected for synchronizing either generator with the bus and the bus with either bus tie circuit.
(k) One local control switch for control of the speed adjusting motor on the governor of the generator prime mover for each local generator.
(ℓ) One transfer switch to transfer the governor motor control from the local control switch to the remote control switch on the master control benchboard. It shall also control the indicator light on the master control unit to indicate when the remote switch is in control. This transfer switch shall be provided only on control benchboards for which the remote control switch, item (m), is to be provided on the master control benchboard.
(m) One remote control switch and indicator light (white) to control the speed adjusting motor on the governor of the remote generator prime mover where the unit has control of a remote generator. The control switch power shall be connected through the transfer switch, item (ℓ), of the remote control benchboard, and the indicating light shall be energized when the transfer of control to this remote control switch is accomplished.
(n) One control switch with indicator light (blue) for control of each local generator circuit breaker.

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(o) One control switch with indicator light (blue) for control of each bus tie circuit breaker.
(p) One control switch with indicator light (blue) for control of each electrically operated distribution circuit breaker.
(q) One control switch with white indicator light for control of generator space heater (when provided with the generator) for each local generator.
(r) Test receptacles with test plugs for connecting portable instruments to the metering current and potential circuits.
(s) One transfer switch with two white indicator lights for connecting the switchboard control bus to the local 450 volt ship service bus, or the 450 volt emergency supply. The transfer may be accomplished automatically, however, in such cases one source of power shall be the local 450 volt ship's service bus and the second source shall be a 450 volt bus tie supply.
(t) One set of three indicator lights with a control switch to indicate ground conditions on the 450 volt bus or any of its connected feeders.
(u) Fuse panel for fusing the control circuits.
(v) One mimic bus system (dark type) simulating the generator and bus tie connections and shall consist of the following: Indication lights (white) for each generator to indicate generator running and up-to-voltage. Indication lights (amber) for each generator and bus tie circuit breaker. Single pole, double throw switches shall be provided for each circuit breaker light. These switches shall be combined with their respective amber lights and so arranged that when the switch handle indication and the circuit breaker position (open or closed) are the same, the light shall be out. When the circuit breaker position and the switch handle indication are different, the light shall be lit. For indicator light power supply, (see 3.18.7). Mimic bus material shall be black laminated plastic 1/8 inch thick in accordance with types NDP or GCP-H of Specification MIL-P-78. Indication of switchboard main and bus tie buses shall be 3/4 inch wide. Generator buses and bus tie cable runs shall be 1/4 inch wide.
(w) One switch for control of the exterior lighting units.
(x) One switch for control of the interior lighting units.
(y) Fluorescent exterior lighting units.
(z) Incandescent interior lighting units.
(aa) One ammeter for reading interchange of current in the bustie.
(bb) One ammeter for each remote generator for reading one phase of generator current.
(cc) One shore power circuit breaker control switch with one blue indicator light to indicate breaker closed. Two white indicator lights, one to indicate shore power feeder energized and the other to indicate the bus tie feeder energized. These two indicator lights shall be mounted near the shore power circuit breaker control switch together with a warning plate stating: "Do not close this circuit breaker when the bus tie circuit is energized."
(dd) One control switch with one blue and one white indicator light for each electrically operated circuit breaker feeding an emergency switchboard.

[1] For units controlling two local generators, each frequency meter, item (h), shall be provided with a transfer switch for reading the frequency of one generator and one bus tie circuit, in which case item (i) shall be omitted.

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3.3.5.2 Submarine control benchboard.- The following devices shall be mounted on submarine control benchboards:

(a) For each ship service turbine generator: items (a), (b), (e), (k), (n) and (q) of 3.3.5.1.
(b) For each ship service motor generator: items (a), (b), (n) and (q) of 3.3.5.1 and the following additional devices:
    (1) Control switches, rheostats, and similar devices required in conjunction with speed regulating equipment.
    (2) One control switch with indicator light (blue) for control of each motor circuit breaker.
    (3) One d.c. ammeter, zero center scale for reading d.c. current.
(c) For a d.c. emergency generator.
    (1) Generator field rheostat.
    (2) One d.c. ammeter for generator field.
    (3) One d.c. ammeter armature.
    (4) One prime mover shutdown switch.
    (5) One control switch with indicator light (blue) for control of the generator circuit breaker. For an a.c. emergency generator: (a), (b), (c), (e), (h), (k), (n) and (q) of 3.3.5.1.
(d) For the battery:
    (1) Two zero center scale d.c. ammeters, one high range and one low range, to measure battery charge and discharge currents.
    (2) One battery float charge measuring device.
    (3) One hydrogen detector remote indicator.
    (4) One control switch with indicator light (blue) for control of each battery circuit breaker.
    (5) One battery exhaust fan control switch with indicator light (green).
    (6) One battery ventilation flow indicator.
    (7) One control switch with ammeter to control and indicate operation of battery recirculating fans.
(e) For the system:
    (1) One synchroscope with synchronizing selector switch for synchronizing generators with the bus and the bus with bus tie circuits.
    (2) Two frequency meters and two voltmeters connected through the synchronizing selector switch for reading frequency and voltage on each side of the point at which synchronizing is being performed.
    (3) One d.c. voltmeter with selector switch for reading battery and d.c. bus voltages.
    (4) One transfer switch with two indicator lights (white) for connecting the a.c. control bus to a 450 volt, normal and alternate source of supply.
    (5) One transfer switch with two indicator lights (white) for connecting the d.c. control bus to a normal and alternate d.c. supply.
    (6) A direct reading ground detector.
    (7) Fuse panel for fusing control circuit.
    (8) One mimic bus system simulating the generators, bus ties, shore power and battery connections of the power system (see 3.3.5.1, item (v)).

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3.3.6 ACB circuit breaker distribution unit.- This unit shall provide for the distribution of power from the main bus. The following appurtenances shall be provided:

(a) Circuit breakers, type ACB, 500 V., 3-pole, manually operated, drawout mounted, having current ratings and protective features as required. 1600 H frame size circuit breakers shall be electrically operated.
(b) Indicator lights (blue) mounted on the front panel of the unit or on the circuit breaker escutcheon to indicate when each of the feeder circuit breakers is closed.
(c) Control switches shall be provided for electrically operated circuit breakers. Where the switchgear group has a control benchboard, these switches shall be located on the control benchboard.

Note.- Where a circuit breaker supplies a feeder which may be energized from another source such as the emergency switchgear group, a white light shall be provided to indicate feeder energized.

3.3.7 AQB circuit breaker distribution unit (see figure 1).- This unit shall provide for the distribution of electric power or lighting supply (450 or 120 V.) through type AQB, 500 V., 3-pole circuit breakers.

3.3.7.1 The number and size of circuit breakers and overall dimensions of the unit shall be in accordance with Drawing 9000-S6201-74453.

3.3.8 ACB - AQB circuit breaker distribution unit.- This unit shall provide for distribution of power through both ACB and AQB type circuit breakers. It shall contain the appurtenances listed in 3.3.6 through 3.3.7.1.

3.3.9 D.c. generator and bus tie unit for a.c. switchgear groups.- This is a special unit not shown on Drawing 9000-S6201-74453. It shall provide for control of a motor-generator set used for ship's service d.c. power and for switching of the d.c. power output. The following appurtenances shall be mounted within the unit:

(a) One generator circuit breaker, type AQB-A250 or type ACB, 250 V., d.c., 2- or 3-pole, manually operated. Type ACB circuit breakers shall be drawout mounted.
(b) One type AQB circuit breaker, 250 V., d.c., 2- or 3-pole, for the bus tie feeder.

3.3.9.1 The following appurtenances shall be mounted on the front of the unit:

(a) One d.c. voltmeter and transfer switch for reading generator voltage and bus tie voltage.
(b) One d.c. ammeter complete with shunt for reading the current of the local generator.
(c) One control switch for use with the motor controller to start and stop the motor of the motor-generator set (to be provided only when the motor controller is not readily accessible).
(d) One indicator light (blue) for indicating the closed positions of the generator circuit breaker.
(e) One set of two ground detector lights and one control switch for indication of a ground on the d.c. bus or its connected feeders.
(f) Space, mounting, and wiring for the generator field rheostat and operating mechanism to be furnished with the generator.

3.4 Emergency power switchgear groups for vessels having a.c. power.- Switchgear groups for controlling larger than 100 kw. generators shall employ a three-way type ABT-A3 (ES) circuit breaker type of Specification MIL-S-17773, bus transfer equipment, and shall be constructed as sectionalized switchgear groups having a separate generator and bus transfer section, and feeder sections. To effect

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a saving of space and weight for switchboards controlling emergency generators rated 100 kw. and less, the bus transfer equipment, type ABT-A3 (ES), contactor type shall be used in lieu of the three-way (type ABT-A3 (ES) circuit breaker type) transfer. When the contactor type unit is used, the switchboard shall be assembled with the AQB distribution units to form a single section (see figures 2 and 3).

3.4.1 A.c. emergency generator control and circuit breaker type bus transfer section. - The following appurtenances shall be mounted on or within the unit:

(a) One type ABT-A3 (ES), circuit breaker type, of Specification MIL-S-17773, 3-way, automatic bus transfer equipment having a current rating not less than the continuous rating of the emergency generator and the current rating of the feeders from the ship's service switchboards, as specified. The associated control switches, indicator lights, and relays suitable for panel mounting shall be mounted on one of the front panels. Circuit breakers shall be drawout mounted.
(b) Necessary disconnecting devices for disconnecting cables between switchgear sections, and performing other switching functions.
(c) Two potential transformers for metering and relaying.
(d) One type AQB circuit breaker 100 amp. frame size, 500 V., 3-pole, instantaneous element only, for power supply to the booster pump supplying cooling (salt) water to the Diesel engine as specified (see 6.1). This circuit breaker shall be connected to the generator bus ahead of the generator circuit breaker. A red warning instruction plate shall be provided stating that the circuit breaker shall always be in the closed position when the generator is set-up for starting.
(e) Two current transformers for metering and relaying.
(f) One fuse panel for fusing the potential transformer primary circuits.
(g) [1]One circuit breaker type AQB or AQB-LF, A250, 500 V., 3-pole, 250 amp. frame size with element rated at 250 amperes. Instantaneous trip setting shall be as follows: AQB circuit breaker 3400 amperes, AQB-LF circuit breaker 3000 amperes. For generators rated 1000 kw. and below and AQB circuit breaker shall be furnished, unless otherwise specified in the contract or order. For generators rated above 1000 kw., an AQB-LF fused circuit breaker shall be furnished. The circuit breaker shall be connected to the generator bus ahead of the generator circuit breaker.
(h) [1]One casualty power terminal, riser type, Drawing 9-S-5302-L, connected through the AQB or AQB-LF circuit breaker, item (g).

[1]Where the rating of the emergency generator is in excess of 400 kw., 2 each of items (g) and (h) shall be furnished.

3.4.1.1 The following devices shall be mounted on the front of the panel:

(a) One a.c. ammeter for reading one phase of generator current.
(b) One a.c. voltmeter for reading one phase of generator potential.
(c) One polyphase wattmeter for reading generator output.
(d) One frequency meter for reading generator frequency.
(e) One control switch with white indicator light for control of the generator space heaters when provided with the generator.
(f) Instruction plates containing complete operating instructions for the automatic equipment and a single line diagram of the main busses.
(g) One set of three indicator lights and switch for ground indication on the 450 V. emergency power bus or any of its connected feeders.

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(h) One indicator light with green color cap and so connected as to indicate when all manually operated control devices are setup for automatic operation. The applicable following functions and conditions shall be included to complete the green light circuit:
  (1) Generator circuit breaker (closed)(when separated from bus transfer).
  (2) Voltage regulator transfer switch (automatic position).
  (3) Ship's service bus transfer equipment (set for automatic operation).
  (4) Emergency bus transfer equipment (set for automatic operation).
  (5) Diesel starting control circuits (set for automatic starting).
  (6) Diesel starting air shut-off valve (open).
  (7) Electrically operated emergency bus unloader switch (if used) (set for automatic operation).

(i) Space mounting and wiring for the voltage regulator equipment, including current and potential transformers, as specified in Specification MIL-R-2729, and for the exciter field rheostat, all equipment to be furnished with the emergency generator. The control switches, rheostats, and similar equipment designed for panel mounting shall be suitably located for front-of-panel operation.

(j) Space mounting and wiring for equipment, as specified in Specification MIL-S-17773, for the control switches and mechanical interlocks, furnished with the bus transfer equipment. The control switches and similar equipment designed for panel mounting shall be suitably located for front-of-panel operation.

3.4.2 A.c. emergency generator control and contactor type bus transfer section. - This unit shall be equivalent functionally to the section specified in 3.4.1, but shall employ type ABT-A3 of Specification MIL-S-17773 bus transfer equipment to provide a unit which is smaller in size. The appurtenances shall be the same as required in 3.4.1, except that the following shall be provided in lieu of 3.4.1 (a):

(a) One generator circuit breaker, type AQB, 250 amp. frame size, 500 V., 3-pole, manually operated, not draw-out mounted.

(b) Two type ABT-A3 (ES), contactor type of Specification MIL-S-17773, bus transfer equipments having current ratings not less than the current rating of the feeders from the ship's service switchboards, as specified. One equipment shall be connected for selection of the ship's service supply, the second equipment shall be connected for transfer between ship's service supply and the emergency generator supply.

(c) Four indicator lights (blue) to indicate closed positions of the contactors of the type ABT-A3 bus transfer equipments. The indicator light circuits shall be wired through auxiliary contacts provided in the bus transfer equipments.

(d) Two indicator lights (white) to indicate that the ship's service power feeders are energized.

(e) One type NQB 500-V., 3-pole circuit breaker, connected to by-pass the bus transfer equipment selecting between ship's service and emergency so that emergency generator power may be supplied to the ship's service switchboards when their generators are secured.

(f) Two control switches, one for each bus transfer equipment, for selecting between manual and automatic control of the equipments.

(g) One electrically operated emergency bus unloader switch complete with control relay and current transformer. This switch shall be connected between the feeder for the steering power switchboard and the emergency bus to disconnect the emergency bus during periods of high current requirements by the steering gear auxiliaries when energized from the emergency generator. The equipment shall function to hold the unloader switch closed unless the demand on the steering power switchboard exceeds 150 percent of the full load current of two steering motors plus 100 percent of the full load current of all other steering gear auxiliaries, in which case the unloader switch shall open provided the power is being supplied by the emergency generator. This item shall be furnished only when specified (see 6.1).

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3.5 Distribution units.- The distribution units of the emergency switchgear group shall be in accordance with 3.3.6 through 3.3.8.

3.6 Load center distribution sections.- These sections shall be switchgear unit type construction. The units comprising the section shall provide for distribution through type ACB or AQB circuit breakers, and shall be constructed in accordance with 3.3.6 through 3.3.8.

3.6.1 The overall dimensions of the unit and the number and size of circuit breakers shall be in accordance with Drawing 9000-S6201-74463. Distribution units receiving two sources of supply shall be provided with interlocked NQB or ACB circuit breakers. Where interlocked circuit breakers are installed, an indicator light shall be provided for each power supply to indicate when power is available.

3.6.2 Each load center distribution section shall be provided with two type AQB-A250 circuit breakers calibrated as required herein. For casualty power supply and two casualty power terminals, riser type conforming to Drawing 9-S-5302-L, connected through the type AQB circuit breakers to the main distribution or power bus.

3.7 Ship's service power switchgear group for d.c. vessels.-

3.7.1 D.c. generator control unit.- The generator control unit shall contain the following appurtenances:

(a) One circuit breaker, type ACB, AQB-A250 or AQB-A400, 250 V., d.c. manually operated, drawout mounted, or AQB, 250 amp., manually operated, not drawout mounted. The number of poles shall be based on the service application (two-wire or three-wire).
(b) One set of disconnecting devices for isolating the circuit breaker from the main bus. (To be required only when casualty power terminals are installed or when the switchboard consists of two or more generator units or one generator unit and one bus tie unit.)
(c) One d.c. voltmeter and switch for reading bus and generator voltage.
(d) One d.c. ammeter complete with shunt for reading generator current. (For three-wire systems, two ammeters shall be provided for reading the current in both the positive and negative generator busses.)
(e) One indicator light (blue) to show the closed position of the generator circuit breaker.
(f) One set of two ground detector lamps and one control switch for indication of ground conditions on d.c. bus or its connected feeders.
(g) Space, mounting and wiring, for the generator field rheostat and operating mechanism to be furnished with the generator.
(h) One reverse current relay to prevent the generator from motoring on the bus after loss of driving power from prime mover. (Provide only where more than one generator is connected to the d.c. bus.)

3.7.1.1 Switchboards which are to be equipped with casualty power connections shall be provided with the following:

(a) One circuit breaker, type AQB-A250, 250 V., d.c., 2-pole, 250 amperes frame size with element rated at 250 amperes (instantaneous trip setting 3800 amperes) for casualty power supply.
(b) Two casualty power terminal blocks (d.c.), riser type, conforming to Drawing 9-S-5288-L, connected through the AQB breaker to the generator bus between the circuit breaker and the disconnecting device.

Note.- The lower section of the unit may be used for the mounting of distribution switches or circuit breakers where the utilization of this space will result in a saving in the overall size of the switchboard.

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3.7.2 D.c. bus tie unit.- The following appurtenances shall be mounted within the unit:

(a) One circuit breaker, type ACB, AQB-A250 or AQB-A400, 250 V., d.c., manually operated, 2 or 3 pole, as required by the service application.
(b) Necessary disconnecting devices for performing switching functions as may be required.
(c) Fuse panel for fusing control circuits.
(d) Shunt for bus tie ammeter (in positive bus only).

3.7.2.1 The following devices shall be mounted on the front panel of the bus tie unit:

(a) One indicator light (blue) to indicate the closed position of the bus tie circuit breaker.
(b) One ammeter with zero center scale for indicating power flow in the bus tie. The ammeter shall be connected to read to the right for power flowing from the bus to the bus tie.
(c) Voltmeter with transfer switch for reading the voltage of the switchboard bus and of the bus tie.

Note.- The lower section of the unit may be used for the mounting of distribution switches or circuit breakers where the utilization of this space will result in a saving in the overall size of the switchboard.

3.7.2.2 The bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus tie and shore power unit". In addition to the equipment specified in 3.7.2 and 3.7.2.1, this unit shall have the following:

(a) One shore power connection circuit breaker, type ACB, or type AQBA-250, 250 V., manually operated, 2 or 3 pole, as required by the application, drawout mounted (type AQB-250 is not drawout mounted), and having a continuous current rating approximately equal to, but not less than, the current rating for the shore connection as specified (see 6.1). The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. The ammeter shunt (see 3.7.2 (d)), shall be so located that it is in the circuit when feeding power to the local switchboard.
(b) One indicator light (blue) to indicate the closed position of the shore connection circuit breaker.
(c) One indicator light (white) to indicate when the shore connection feeder is energized.
(d) One indicator light (white) to indicate when the bus tie feeder is energized.
(e) Warning plate adjacent to the indicator lights for the shore connection and bus tie feeder to warn against closing the shore connection circuit breaker when the bus tie is energized.

3.7.2.3 For installations not requiring a bus tie unit, items (a), (b) and (c) of 3.7.2.2, shall be mounted on one of the ACB distribution circuit breaker units and the shore power circuit connected to the switchboard bus. In such instances, this unit shall be designated as "ACB distribution circuit breaker and shore power unit", and the following additional equipment shall be provided:

(a) One indicator light (white) to indicate when the switchboard bus is energized.
(b) Warning plate adjacent to the indicator lights to warn against closing the shore connection circuit breaker when the switchboard bus is energized.

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3.7.3 D.c., ACB circuit breaker distribution unit.- The unit shall provide distribution feeder circuit breakers for power from the main bus and shall be in accordance with Drawing 9000-S6201-74453. The following appurtenances shall be provided:

(a) Circuit breakers, type ACB, 250 V., d.c., manually operated, draw-out mounted, 2 or 3 pole, as required by the service application.
(b) Indicator lights (blue) mounted on the front of the unit to indicate when the circuit breakers are closed.

3.7.4 D.c., AQB circuit breaker distribution unit.- This unit shall provide for the distribution of power or lighting through type AQB, 250 V., d.c. circuit breakers.

3.7.4.1 The number and size of circuit breakers and overall dimensions of the unit shall be in accordance with Drawing 9000-S6201-74453.

3.8 D.c. emergency power switchgear group.- The d.c. emergency switchgear group shall be functionally equivalent to the emergency switchgear group for a.c. vessels as specified in 3.4 and shall employ the equivalent type units as required.

3.9 Battery charging and discharging switchboards.- When special units not delineated on Drawing 9-S-5370-L are required, their design shall conform to the following:

3.9.1 Methods of calculating the resistance and current carrying capacity of resistors and rheostats for battery charging are as follows:

3.9.1.1 For two step, fixed resistor charging, use formulas (a), (b), (c), and (d).

(a) $R_1 = \frac{110-2.35N}{I_1}$

(b) $C_1 = \frac{120-2.1N}{R_1}$

(c) $R_f = \frac{110-2.5N}{I_f}$

(d) $C_f = \frac{120-2.2N}{R_f}$

3.9.1.2 For single step, fixed resistor charging, use formulas (c) and (d) of 3.9.1.1, except substitute 2.1 for 2.2 in formula (d).

3.9.1.3 For variable rate, rheostat charging and discharging, use formulas (e) and (f) and (g) of 3.9.3.

(e) $R_t = \frac{120-2.2N \text{ min.}}{I_f \text{ min.}}$

(f) $R_d = \frac{1.75N \text{ min.}}{I_d \text{ max.}}$

3.9.2 Values of total resistance of the successive steps on the rheostat shall approximate a logarithmic scale.

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3.9.3 Current-carrying capacity of the rheostat shall be tapered so that the current-carrying capacity of each step on the rheostat shall not be less than,

(g) $C_S = \frac{120\text{-}2.1N \text{ min.}}{R_S}$

but need not be greater than I_i max. + 10 percent of I_d max. + 10 percent, whichever is the greater.

R_i = resistance, in ohms, to be connected in series with the storage battery to obtain I_i amperes at the end of the initial charge.

R_f = resistance, in ohms, to be connected in series with the storage battery to obtain I_f amperes at the end of the finish charge.

R_d = maximum resistance, in ohms, for the first step on the rheostat to be connected in series with the minimum number of the largest ampere hour capacity cells to be discharged to obtain a discharge current in amperes of I_d.

R_t = total resistance, in ohms, to be connected in series with the minimum number of cells to be charged to obtain I_f min. amperes. R_t may be increased to allow for manufacturing tolerances.

R_S = sum of resistances, in ohms, of each successive step (in use) on the rheostat.

I_i = recommended initial charging rate, in amperes.

I_i max. = recommended initial charging rate, in amperes, of the largest ampered-hour capacity battery to be charged.

I_f = recommended finish charging rate, in amperes.

I_f min. = recommended finishing charging rate, in amperes, of the smallest ampered-hour capacity battery to be charged.

I_d = recommended discharge rate, in amperes (see table I).

I_d max. = highest recommended discharge rate in amperes (taken from table I) for which variable rheostat is to be used.

C_i = minimum current-carrying capacity, in amperes, of the initial charging resistor.

C_f = minimum current-carrying capacity, in amperes, of the finish charging resistor.

C_S = minimum current-carrying capacity, in amperes, of each successive step on the rheostat.

N = number of cells to be charged in series, which shall not be greater than 36 for switchboards operating from a 120 V., d.c. supply.

N min. = minimum number of cells to be charged by the circuit.

N max. = maximum number of cells to be charged by the circuit.

1.75 = lowest counter e.m.f. per cell that may be expected during discharging.

2.1 = lowest counter e.m.f. per cell that may be expected during charging.

2.35 = average counter e.m.f. per cell just before the end of initial charge.

2.2 = the average counter e.m.f. per cell at the start of the final charge.

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2.5 = the average counter e.m.f. per cell just before the end of the finish charge.
110 = minimum voltage at the battery charging switchboard.
120 = maximum voltage at the battery charging switchboard.

Note. - Application for special voltages shall be calculated on a similar basis.

Table I - Battery charging and discharging rates in amperes.

Ampere hour rating	Charging rates		10 hour discharge rate
	Start	Finish	
Bureau of Ships batteries			
15	1.5	1	1.5
20	2	1	2
60	8	4	6
100	14	7	10
130	30	15	13
175	30	15	17.5
200	32	16	20
205	44	22	20.5
270	44	22	27
300	44	22	30
Bureau of Aeronautics batteries			5 hour discharge rate
11	2	1	2.2
17	3	1.5	3.4
34	6	3	6.8

3.10 Test switchboards. -

3.10.1 Special test switchboards required for applications where a standard unit cannot be used, shall provide the following features, together with the functional requirements of the application:

3.10.1.1 Safety features. - Completely dead-front construction shall be employed to the greatest extent practical. Where live parts such as fuse test contacts must be exposed, these parts shall be energized at the lowest practicable potential, usually 6 V. All exposed metal parts of the unit shall be grounded.

3.10.1.2 The unit shall be suitable for mounting on a work bench.

3.10.1.3 An instruction plate containing a complete diagram of connections and operating instructions shall be provided and mounted on the front of the unit, if practical. When such mounting is not practical, this instruction plate shall be provided for mounting adjacent to the unit.

3.11 Miscellaneous switchboards. - Switchboards for special purposes, such as arc welding generator control, electric furnace control, ship's alongside power and degaussing control, which are not described in detail, ~~shall be as specified~~ (see 6.1) as regards functional requirements and selection of appurtenances. The design and construction shall conform to this specification.

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3.12 Switchboard structure and its compartmentation. -

3.12.1 General construction. -

3.12.1.1 Standardized switchgear sections. - Each unit of a switchgear section shall be in accordance with the applicable unit type shown on Drawing 9000-S6201-74453 and applicable notes on this drawing. The units shall be constructed with a self-supporting, boxlike framework for support of the front panels, buswork, circuit breakers, and other appurtenances to be included in the various units which constitute the section. Each edge of each unit shall be formed by a continuous length of structural, shape, formed member, or formed members welded to effect a continuous length. Units shall be bolted together to form sections, and shall be connected by transverse busses. Mounting and inter-unit bolt hole locations and bus locations shall be as shown on Drawing 9000-S6201-74453, except that where bus locations are not so specified, they shall be installed to suit the application. The framework of each section shall be constructed so that under conditions of class HI mechanical shock specified in Specification MIL-S-901, the members will not be damaged and any distortion will not be of sufficient magnitude to impair the operation of the equipment contained within the section. Limitations of weight and space required for all apparatus for installation on Naval ships dictate that switchboards be as small and light as practicable. The materials and spacing employed in the construction of switchboards shall be limited in all cases to that required to meet the specification requirements, in order that excess weight and size will be avoided.

3.12.1.2 Special switchgear sections. - Each section of a switchgear group shall be constructed with a rigid, self-supporting, boxlike framework for support of the front panels, buswork, circuit breakers, and other appurtenances to be included in the various units which constitute the section. Each edge of each unit shall be formed by a continuous length of structural shape, formed member or formed members welded to effect a continuous length. The framework for a section may be constructed as an integral frame for all units, using common vertical and horizontal members at the junction of adjacent units, or may be formed by constructing an independent box framework for each unit joining the individual units by bolting or welding to form the section. Where each edge of the unit is formed by welding formed members, the ends joining to form a section shall be welded including necessary intermediate welding of the joining members to add strength and rigidity to the section. The framework of each section shall be constructed so that under conditions of class HI mechanical shock specified in Specification MIL-S-901, the members will not be damaged and any distortion will not be of sufficient magnitude to impair the operation of the equipment contained within the section. Limitations of weight and space required for all apparatus for installation on Naval ships dictate that switchboards be as small and light as practicable. The materials and spacing employed in the construction of switchboards shall be limited in all cases to that required to meet the specification requirements in order that excess weight and size will be avoided.

3.12.2 Dimensions. - Insofar as practicable, the grouping of the various units which comprise a switchgear group shall be such that the length of each section is approximately equal to the height; however, the length of a section shall not exceed approximately 8 feet. For sections less than 6 feet in length, special consideration shall be given in the design of the frame to provide sufficient stiffness in the lengthwise direction. The maximum width of each special unit shall not exceed 34 inches, except where a single large capacity circuit breaker requires a greater width. Width of standardized units shall be as specified on Drawing 9000-S6201-74453; however, the width of the unit shall be adequate to allow removal of the circuit breakers through the opening provided by the hinged front panel. In the arrangement of the appurtenances to be located within the structure, and the general design of the unit, particular attention shall be given to limiting the height and keeping the center of gravity of the unit as low as practicable, consistent with good design and facility of operation.

3.12.3 Framework fabrication. -

3.12.3.1 Standardized units. - The framework shall be fabricated from angles, channels, or other structural shapes, or formed members. Each edge of each unit shall be formed by a length of a structural shape, formed member or formed members welded to effect a continuous length. Intermediate vertical supporting members shall be provided for the support of such heavy appurtenances as circuit breakers and large disconnecting switches. Gusset plates shall be welded at the corners of the units in both lengthwise and lateral vertical planes such that each corner shall have a strength not less than the bending strength of the weakest adjoining vertical or horizontal member. Typical bottom corner construction is shown on figure 4.

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3.12.3.2 Special sections.- The framework shall be fabricated from angles, channels, other structural shapes, or formed members. Each edge of each section shall be formed by a length of a structural shape, formed member or formed members welded to effect a continuous length. Intermediate vertical supporting members shall be provided for the support of such heavy appurtenances as circuit breakers and large disconnecting switches. Gusset plates shall be welded at the corners of the units in both lengthwise and lateral vertical planes such that each corner shall have a strength not less than the bending strength of the weakest adjoining vertical or horizontal member. Typical bottom corner construction is shown on figure 4.

3.12.4 Gusset plates.- In the special case where the lengthwise gusset plates at the front of the unit interfere with appurtenances, special consideration shall be given to the design of the corner to provide maximum strength. Small gusset plates or L-shaped plates may be used in the above special case. Gusset plates shall be not less than 1/8 inch thick; however, where the thickness of the gusset is less than 3/16 inch, the diagonal edge shall be bent to form a flange. Three sided, forged or formed and heat treated aluminum alloy gussets maybe used on aluminum constructed switchboards. Typical gusset plates are shown on figure 5.

3.12.5 Framework assembly.- The framework of each standardized unit or special section shall be stiffened to resist bending by the use of diagonal braces or welded sheets, but braces and sheets shall not be secured in such manner as to restrict access to the front and rear of the units. The framework of each unit or section shall be fabricated by welding all members except those which must be disassembled for replacement of appurtenances. In these cases bolting shall be employed. Riveting shall not be used in the assembly. Interframe bolting or standardized units to form sections shall be in locations as shown on Drawing 9000-S6201-74453. The minimum size of bolt used for this purpose shall be 1/2 inch diameter. The structural shapes that comprise the base members shall not be less than 1/4 inch thick and the horizontal flanges which are bolted to the foundation shall not be less than 2 inches wide.

3.12.6 Bolting to foundation channel.-

3.12.6.1 Standardized units.- Bolting to foundation channel shall be as shown on Drawing 9000-S6201-74453. Where the thickness of the frame member is less than 3/8 inch, a 1/4 inch thick reinforcing washer, approximately 2 inches square, shall be provided under each bolt head.

3.12.6.2 Special sections.- The frame of each section shall be bolted directly to the channel by 5/8 inch or 3/4 inch diameter bolts conforming to Specification MIL-B-857. The minimum number of holding-down bolts to be provided for a section shall be based on the total weight of the section, allowing 140 pounds for each 5/8 inch diameter bolt or 210 pounds for each 3/4 inch diameter bolt. Not less than 4 bolts shall be provided per unit. The holding-down bolts shall be located as near the vertical frame member as practicable, but with not more than 3 inches horizontal separation between the vertical member and the center of the bolt hole. Where more than 12 bolts are required, two in each corner, and two adjacent each intermediate vertical member, the additional bolts shall be located an near the corner bolts as practicable. Where the thickness of the frame member is less than 3/8 inch, a 1/4 inch thick reinforcing washer, approximately 2 inches square, shall be provided under each bolt head.

3.12.7 Side sheets.- The ends of each standardized unit or special section shall be enclosed by solid sheets, at least 1/16 inch thick, welded to the switchboard framework, except where the arrangement of switchgear within the section is such as to require side access for maintenance, in which case the side sheet or a portion thereof shall be removable. Removable side or rear sheets are not considered to be structural members and the tests required shall be made with such sheets removed. The side sheets shall be welded by intermittent welds or spot welds to all vertical and horizontal frame members. The gusset plates in the plane of the sheet may be omitted, provided the sheet is welded to the frame at the corners with the same length of welding that would be normally employed with a gusset plate.

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3.12.8 Compartmentation. -

3.12.8.1 Solid sheets shall be provided between adjacent units of a section to add strength and stiffness to the structure and to minimize the possibility of spread of fire. These sheets shall be cut away to provide for interconnecting bus bars and wiring, and suitable insulating spacers shall be employed to close the openings. The intermediate sheets shall be welded to the framework of at least one of the two adjacent units. In those cases where a multiplicity of interconnections exist, this requirement for intermediate sheets may be waived.

3.12.8.2 Each unit shall be internally subdivided by sheet metal barriers to form separate compartments for the circuit breakers to isolate them from the bus work and cable connections. When barriers are installed insulation distances shall be as specified in 3.16. A separate compartment shall be provided for each type ACB circuit breaker, but all type AQB circuit breakers on one unit may be in a common compartment. The compartments shall be constructed sufficiently tight to prevent an arc or fault from spreading from one compartment to another or into the area occupied by the bus work. Bus disconnecting devices provided with fixed mounted generator or bus tie circuit breakers shall be located outside the circuit breaker compartment. When the rear enclosure of the AQB circuit breaker compartment is used for support of the circuit breakers, the terminal mounting block shall be bolted through reinforcing angles or channels secured to the sheet enclosure. Each type ACB circuit breaker compartment shall be enclosed on the front by a separate hinged panel. The type AQB circuit breaker compartments shall be enclosed on the front by a removable panel section. On type AQB circuit breaker compartments over 60 inches high, the front enclosure shall be made by two panels.

3.12.9 Ventilation. - The compartments shall be provided with adequate ventilation for cooling of equipment. In providing means for ventilation, consideration shall be given the fact that the switchboards may be operated under conditions favorable to condensation of water vapor within the unit. Ventilation for type ACB circuit breakers shall be arranged so that ionized gases cannot be transmitted from one compartment to another. Ventilation for type AQB circuit breaker compartments shall be provided by allowing air to enter the compartment below the circuit breaker mounting sheets and escaping from the compartment through an opening in the front enclosing panel above the top row of circuit breakers. In any case where the bottom of the circuit breaker compartment is higher than the base of the unit, the rear sheet shall not be punctured, but openings shall be provided in the front panel for entrance of air. The area of the openings at the top and the bottom of the compartment shall be based on allowing not less than 6 square inches of opening for each 100 AMP frame size, and 10 square inches for each 225 or 600 AMP frame size circuit breaker, using the maximum requirement for any horizontal row of circuit breakers in the compartment. Ventilating louvers and grill openings shall be designed to minimize probability of entrance of solids and shall protect personnel against injury from electric arcs.

3.12.10 Structural rigidity. - Units of all switchgear groups, except test switchboards, shall be designed and constructed so that when mounted on a rigid foundation all natural frequencies in bending shall be greater than 23 cycles per second. The natural frequencies of vibration shall be tested by methods specified in 4.3.2. The members in the top of the frame of all switchboards shall have sufficient strength and rigidity for securing top braces. The braces shall not be considered as a part of the switchboard.

3.12.11 Materials. - The switchboard structure may be either of steel or aluminum construction as specified (see 6.1). Steel conforming to Specification QQ-S-692 or QQ-S-693 shall be utilized in the fabrication of switchboard structure, panels, front, side, top, and back sheets. Aluminum alloy conforming to temper T6 of Specification QQ-A-325 or temper T6 of Specification QQ-A-270 shall be used for structural pieces. Aluminum alloy conforming to temper H32 or H34 of Specification QQ-A-318 shall be used for plates and sheets. Aluminum alloy conforming to temper H22 and H24 of Specification QQ-A-359 may be used for nonstructural applications.

3.12.12 Thread locking devices. - Locking devices shall be provided on the following basis for electrical connections and mechanical assemblies in all switchboards and switchgear units, and in all equipment unless otherwise specified in the individual equipment specifications.

3.12.12.1 Electrical connections. - All nuts, bolts, studs, and screws used for electrical connections shall be secured by means of an acceptable locking device, except that a locking device need not be provided where solderless type connectors are used for conductors below 4000 cm.

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3.12.12.2 Mechanical assemblies.- All nuts, bolts, studs, and screws used for structural purposes shall be secured by means of an acceptable locking device where continuous satisfactory operation, under the shock, vibration, and heat conditions specified herein, or as specified in the individual equipment specifications, depends on maintaining tight connection of parts, or where a holding screw, nut, bolt, or component part could fall into the equipment.

3.12.12.3 Acceptable locking devices.- The following are considered acceptable types of locking devices:

(a) Nut and lock nut.
(b) Self locking nut.
(c) Castellated nut with cotter pinning or safety wiring.
(d) Toothed lock washer.
(e) Split ring lock washer.

3.12.13 Welding.- Welding and allied processes shall be in accordance with Standard MIL-STD-278.

3.13 Insulating panels, bases, and bus supports.-

3.13.1 Insulating material.- In order to restrict the use of material which might catch fire or emit fumes when exposed to flame, the quantity of insulating material shall be kept to a minimum and used only where necessary to provide electrical insulation between live parts and to ground. Insulating material for bus supports and spacers, fuse bases, and other similar parts shall be of type GMG, Specification MIL-P-15037 or type GMM of Specification MIL-P-17721.

3.13.2 Panel edges shall be smooth cut to present a neat appearance. Front panels such as used on a live front type switchboard shall have a bevel of not less than 1/8 by 1/8 inch around the front edges. Bases for mounting circuit breakers, contactors, fuses, and other appurtenances located on the rear of the switchboard or within the frame assembly need not be beveled.

3.13.3 The design, number, spacing, and mounting of the bus supports shall be in accordance with calculations based on the magnitude of the short circuit stresses which may occur on the buses being supported.

3.14 Hinged and removable enclosing panels.-

3.14.1 Structure.- The panels shall be made from sheet 3/32 inch to 1/8-inch thick or U.S. Gage number 11. (Commercial tolerances acceptable.) The panel shall be formed with bent angle or channel edges not less than 1 inch deep. The flanges of all panels shall be welded at the corners and ground smooth. There shall be no butt joints on the panel surfaces. Where the strength of the panel is materially weakened by cutouts, it shall be reinforced by suitable stiffening members. The front enclosure panels for all type AQB and NQB feeder circuit breakers shall be drilled with 5/16-inch diameter holes so that the contact prong of the megger testing equipment may be inserted to make contact with the circuit breaker terminal for making the insulation tests of the feeder cables. Type NQB and AQB circuit breakers that are interlocked to form a bus transfer switch shall not be provided with megger holes. The size of the cutout for type AQB circuit breaker operating handles and escutcheons shall be in accordance with Specification MIL-C-17361.

3.14.2 Panel hinging.- Panels shall be hinged where necessary to provide access to the back of the equipment mounted thereon and to equipment mounted within the switchboard frame assembly. Hinged panel sections shall be provided in front of the type ACB circuit breakers for access from the front of the switchboard. Means shall be provided so that these hinged sections may be opened without removing the hand rail. Hinged panels shall be provided with door stops or positioning devices to stop the door in its opening swing, and to hold it in the open position. Panels shall be hinged on the left-hand side when facing front of the panel. A typical panel hinge assembly is shown on figure 6.

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3.15 Mounting of equipment. -

3.15.1 Securing of equipment. - The equipment shall be adequately secured to the panels or framework so as to insure against dislodgment due to shock. Careful consideration shall be given to the size and number of bolts used. Brackets and mounting lugs or feet shall have fully enclosed holes or slots. Open end slots will not be acceptable. All mounting bolts shall be secured by locking devices approved by the bureau or agency concerned.

3.15.2 Mounting bolts and screws. - The bolts used in the assembly of the framework and for securing the large appurtenances to the framework shall be hexagon head type. Screws for miscellaneous purposes shall be round fillister or flat slotted head type. For securing identification plates or other small appurtenances to the front panels; however, consideration will be given to use of special types of heads if their use will facilitate assembly.

3.15.3 Arrangement of equipment. - No apparatus shall extend beyond the edges of the panel on which it is mounted. Symmetrical arrangements of equipment on the units is desirable, but should not take precedence over arrangements where space may be saved for future or spare equipment. Particular attention regarding this matter shall be given in regard to the mounting of circuit breakers.

3.15.4 Mounting of switches. - Single throw switches shall generally be vertically mounted with hinge at the bottom and all double throw switches shall be horizontally mounted. Where necessary to mount single throw switches horizontally, they shall hinge on the right-hand side and throw to the left.

3.15.5 Polarity. - For direct current, like polarities shall be similarly disposed; that is, facing the front of the switchboard or panels, all left-hand contacts shall be of the same polarity (negative) and all right-hand contacts of the same polarity (positive). The upper blades of all horizontally mounted switches shall be of positive polarity. In special cases, in order to save copper or space, transposition of polarities may be desirable. Such cases will require specific approval of the bureau or agency concerned.

3.15.6 Phase rotation. - For alternating current, like phases shall be similarly disposed, that is, facing the front of the switchboard or panel, the phase rotation shall be A, B, and C, respectively, from right to left, top to bottom, or front to back. Consideration will be given to other arrangements, where they afford copper reduction, improved electrical clearances, or space saving. Such cases will be subject to approval by the bureau or agency concerned.

3.15.7 Casualty power equipment. -

3.15.7.1 Arrangements. - The casualty power connection equipment is intended to provide a means whereby temporary circuits may be quickly and readily arranged for distributing power to vital equipment under casualty conditions. This function requires that the casualty power equipment be available for use in the event of damage to the ship's service and emergency distribution systems. For ship's service and emergency switchgear groups, the casualty power circuit breaker and riser terminal (required herein on the generator units) must be available for use in the event of damage to the remainder of the switchboard on which the generator unit is located. It is, therefore, essential that the lead from the generator bus on generator units (or panel bus on steering power transfer switchboard and power distribution units) to the casualty power connection equipment be as short and direct as practicable, and that the casualty power terminal be readily accessible to permit rapid connection of the casualty power cable.

3.15.7.2 Mounting for ship's service and emergency switchgear groups. - For ship's service and emergency switchgear groups, the casualty power circuit breaker and riser terminal shall be mounted on the generator units, and may be mounted at the rear or on the side of the unit. When mounted within the unit, the casualty power equipment shall be readily accessible from the rear of the unit so that cables may be connected to the terminal without undue exposure of personnel to live parts. The rear enclosures shall not be used for support of the terminal or circuit breaker, nor shall its removal be necessary for access to the terminal or for operation of the circuit breaker. When mounted on the side of the unit, the riser terminal shall be surface mounted and the circuit breaker shall be removable without removal of the side sheet. Normally, the circuit breaker and riser terminal shall be located adjacent to one another so that personnel may readily determine that the breaker is open and the terminal not energized before making connections. Where adjacent locations are not practicable, the terminal shall be enclosed to avoid undue exposure of

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personnel to live parts. Tools for riser terminals shall be mounted in a rack on the outside of the switchboard enclosure adjacent to the terminal. In no instance shall the tools be mounted inside the switchboard enclosure. Consideration shall be given to the location of the casualty power connection equipment (circuit breaker and riser terminal) in the arrangement of the units in the switchboard section.

3.15.7.3 Mounting for AQB distribution circuit breaker units.- For AQB distribution circuit breaker units (where casualty power terminals are required), the casualty power terminal may be mounted at the rear, or on the side of the unit, whichever location is more accessible for connection of the incoming cable. The considerations outlined in 3.15.7.2, regarding mounting of the terminal and tools for the terminal, shall apply. The terminal shall be enclosed to avoid undue exposure of personnel to live parts.

3.15.8 Voltage regulator equipment.- The space and arrangement of voltage regulator equipment shall be based on permitting replacement, with the minimum modification of mounting and arrangements of voltage regulators of one manufacturer's type with those of another manufacturer's type, or direct acting rheostatic type with rotary amplifier type regulators. (See Specification MIL-R-2729.) This basis for arrangement and space factors shall be adhered to rigidly.

3.16 Insulation distances.- Insulation distances in air, and surface creepage distances on insulating materials for all appurtenances shall conform with the applicable specifications. For all other live parts, such as bus work, the values shall conform to table II.

Table II - Switchboards.

Voltage	Insulation distances to ground or to opposite polarity		
	In air	Surface creepage	
		Bottom and side surfaces	Top-flat surfaces [1]
	Inches	Inches	Inches
125 a.c. or d.c.	1/2	15/16	1- 1/8
230 a.c.	5/8	1- 3/16	1- 1/2
500 a.c. or d.c.	25/32	1- 9/16	2
1000 a.c.	1- 5/16	2- 1/2	3-11/32

[1] Values for top creepage apply to flat surfaces or to curved surfaces presenting sufficient irregularities to permit accumulation of dust and moisture. Values are not intended to apply to simple cylindrical surfaces having a radius of 3 inches or less where side creepage values will apply.

3.17 Bus bars and connectors.-

3.17.1 Bus connections.- All bus connections shall be made with solid copper or aluminum alloy bus bars except as specified for certain switch and circuit breaker connections.

3.17.2 Copper bus bars.- Copper bus bars shall be in accordance with Specification QQ-C-504, hard-temper, except that the edge of the bus bars shall have a radius of approximately one-half the bar thickness. The bars shall be silver surfaced at the contact areas.

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3.17.3 Cable connector sizes.- On switchboards having an a.c. bus structure designed for short circuit currents of 15,000 AMP or less, single conductor insulated copper cable, type SHFS 42, of Specification MIL-C-915, shall be used for connections to 100 AMP frame size type AQB circuit breakers. On switchboard buses having a total connected d.c. generator capacity of not more than 150 kw. at 120 V., similar cable connections shall be provided for 100 AMP frame circuit breakers and 30, 60, and 100 AMP knife switches. On bus work having a connected d.c. generator capacity of not more than 90 kw. SHFS-9 of Specification MIL-C-915, cable may be used for 30 AMP switches and SHFS-14 for 60 AMP switches. When approved by the bureau or agency concerned, other cable connections may also be used where a definite advantage in construction, accessibility or arrangement may be gained.

3.17.4 Aluminum bus bars.- Rectangular aluminum alloy 6101, T61 temper with rounded corners in accordance with Specification MIL-B-19780 shall be utilized for bus bars. The bars shall be silver-surfaced at the contact areas as specified herein.

3.17.5 Cable connector sizes for casualty power.-

3.17.5.1 Cable connecting a casualty power circuit breaker to the line side of an a.c. generator circuit breaker shall be type SHFS-100 of Specification MIL-C-915. Cable connecting the casualty power circuit breaker and the riser terminal shall be type SHFS-42.

3.17.5.2 Cable connecting a casualty power circuit breaker to the load side of a d.c. generator circuit breaker shall be type SHFS. Cable connecting the casualty power circuit breaker and the riser terminal shall be type SHFS of Specification MIL-C-915. The cable size shall correspond to the element rating of the selected casualty power circuit breaker, based on a SHFS-100 cable being used with a 225 ampere circuit breaker element.

3.17.6 Bus bar sizes.- The size of the bus bars shall be selected on a basis of the current carrying ratings shown in table III, and kept to a minimum as far as practicable. In no case, shall the bus bar sizes be smaller for a given current than the corresponding sizes listed in table III. The ratings shown are based on a 35° C. rise above a 50° C. ambient and on the separation or arrangement of phases sufficient to neglect mutual inductance between phases and interference with radiation of neighboring conductors. If the spacing is not large enough to neglect mutual inductance between phases or interference in radiation, a reduction in the values given in table III shall be made.

Table III - Ampere rating of rectangular bus bars placed on edge.

Number of bars in parallel	Size of bars (inches)	Cross sectional area, (square inches)	A.c. ampere rating 60 cycle		D.c. ampere rating
			Copper bus, silver surface	Aluminum bus silver surfaced	Copper bus, silver surface
1	3/4 x 1/8	0.094	210	140	210
	1 x 1/8	.125	285	190	285
	1-1/2 x 1/8	.188	425	280	425
	2 x 1/8	.250	555	370	555
	3/4 x 3/16	.140	265	175	265
	1 x 3/16	.188	355	235	355
	1-1/2 x 3/16	.278	550	360	550
	2 x 3/16	.375	700	450	710
	3/4 x 1/4	.188	295	200	295
	1 x 1/4	.250	410	300	410
	1-1/2 x 1/4	.375	600	420	600
	2 x 1/4	.500	780	550	800
	2-1/2 x 1/4	.625	1,000	660	1,050

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Table III - Ampere rating of rectangular bus bars placed on edge (cont'd.).

Number of bars in parallel	Size of bars (inches)	Cross sectional area, (square inches)	A.c. ampere rating 60 cycle		D.c. ampere rating
			Copper bus, silver surface	Aluminum bus silver surfaced	Copper bus, silver surface
	3 x 1/4	.750	1,140	800	1,185
	4 x 1/4	1.000	1,425	1,000	1,490
	5 x 1/4	1.250	1,760	1,250	1,850
	6 x 1/4	1.500	2,100	1,500	2,190
2 (1/4 inch apart)	2 x 1/4	1.000	1,350	900	1,370
	3 x 1/4	1.500	1,825	1,250	2,000
	4 x 1/4	2.000	2,280	1,600	2,530
	5 x 1/4	2.500	2,740	1,900	3,100
	6 x 1/4	3.000	3,140	2,250	3,630
3 (1/4 inch apart)	3 x 1/4	2.250	2,200	1,600	2,620
	4 x 1/4	3.000	2,660	2,000	3,110
	5 x 1/4	3.750	3,200	2,400	3,830
	6 x 1/4	4.500	3,600	2,800	4,560
4 (1/4 inch apart)	3 x 1/4	3.000	2,650	1,240	3,130
	4 x 1/4	4.000	3,020	1,568	3,870
	5 x 1/4	5.000	3,450	3,200	4,750
	6 x 1/4	6.000	3,830	4,000	5,550

3.17.7 Grouping and arrangement of bus bars. - Particular attention shall be given to the grouping and arrangement of buses for a.c. (60-cycle and above) currents near magnetic material, since proximity of such material will create additional heating, thereby interfering with the dissipation of heat from conductors and reducing their capacity. Where the available space permits such construction, laminated bus bars may be separated into two groups with an increased spacing between each group, thereby obtaining an increase in current rating without increase in bus bar size. The increase in rating will depend upon the arrangement and spacing of groups.

3.17.8 Bus bar stresses. - The bus work assembly shall be designed to withstand the stresses resulting from the maximum short-circuit currents which might be encountered under normal operation of the electric plant. Conditions of high impact shock which may be encountered aboard the ship shall not mechanically damage the bus work nor reduce the bus clearances below minimum requirements. Where single conductor cable is used for connection to circuit breaker or knife switch, the cables shall be securely bound together with noninflammable cord and secured to the supports as necessary to prevent distortion under short circuit conditions. The binding shall be applied in such a manner that it will not cut into the cable insulation due to the forces exerted under short circuit conditions. The binding shall be especially heavy where cables turn out of the pack. Terminal lugs shall be used for terminating cable at the switch terminals and at the bus bar. Through-bolts shall be used for securing the cable terminal to the bus bar. Not more than two cable terminals shall be clamped by one bolt, and when two are used, they shall be placed on opposite sides of the bus. The cables shall be neatly formed, and where their length exceeds approximately 12 inches, they shall be supported by the switchboard structure.

3.17.9 Bus bar forming. - The bus bars shall be accurately formed and the holes shall be accurately located so as to insure that the bus bars can be fitted into place without being forced. Flat bends shall have an inside radius of not less than the thickness of the bus bar, and the ends of the bus bars shall be neatly finished. Edgewise bends of bars up to

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2 inches in width, may be made on an inside radius of 1 inch, and those over 2 inches, but not exceeding 4 inches in width, may be made on an inside radius of 2 inches. Bars of greater width should not be bend edgewise. Bars shall be free from cracks or flaws at bends.

3.17.10 Silver surfacing of buses. -

3.17.10.1 The silver surfacing of copper buses and connections shall be accomplished by plating or by amalgamation process using 99.9+ percent pure silver. The silver thus applied shall not be less than 0.0005 inch thick at the time of assembly of the bus work, and shall show no tendency to peel. The contact surfaces shall be cleaned to remove all foreign materials immediately prior to assembly. The area to be covered shall, in the case of flat surfaces, be not less than the area covered by the smaller bus bar or connector or an area around each bolt hole equal to not less than that of a 1/8 inch wide band. Threaded surfaces shall have silver thickness of at least 0.0004 inch. In the case of contacts which engage with sliding members, the thickness of silver applied in the above manner shall show no tendency to peel under normal conditions of usage, nor shall the copper show through the silver surface after 10,000 normal operations. The current density and pressure of silver contacts shall be such that the temperature rise of the contact will not increase the temperature rise of the bus bar.

3.17.10.2 The silver surfacing of aluminum buses and connections shall be accomplished by means of a plating process satisfactory to the bureau or agency concerned using 99.9+ percent pure silver. Plating shall not be less than 0.0005 inch thickness, and shall satisfactorily withstand adhesion tests specified in Specification QQ-S-365. The contact surfaces of aluminum bus bars shall be silvered up to 1 inch past the joint area. The entire bus may be silver surfaced at the discretion of the manufacturer.

3.17.10.3 Holes in bus bars for bolting may be either punched or drilled, but the contact area shall be smoothed by belt sanding, or other equivalent method, prior to silver surfacing. Where punching is used, the operation shall not appreciably indent the surface of the bus bar in the vicinity of the hole. Slotted holes are not permitted. The contact surfaces of bus bars and contact nuts shall be finished true to give adequate contact. Contact nuts and contact spacer nuts for copper stud contacts shall be of rolled brass and shall be of sufficient size to provide the necessary area for contact surface. All contact nuts and contact spacer nuts for silvered contacts shall be silver-surfaced, as specified in 3.17.10.1, for threaded surfaces.

3.17.10.4 The joints of all bus bars shall be specially prepared before bolting. The method of preparation shall be as follows:

(a) Apply a thick film of high viscosity paste conforming to Specification MIL-C-16173.
(b) Bolt joints together without removing the paste.
(c) After joints are bolted together, remove paste from all exposed surfaces with a petroleum solvent.

3.17.11 Bolting of bus bars. - Bolts and nuts used in bus bar joints shall be of steel coarse thread series, heat treated, and suitably plated to protect against corrosion. Flat washers shall be of similarly plated steel and shall be used under all bolt heads and nuts adjacent to the conductor. Securing bolts shall be fitted with nuts conforming to figure 7. For typical bus bar mountings see figure 8. Locking devices listed in 3.12.12.3 (b), (d), and (e) are not to be used in bus bar joints.

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3.17.12 Contact pressure of bus bars. - Bus bar joints shall be made in accordance with table IV and figure 9. Bolts for bus bar joints shall be lubricated and the following minimum torques shall be used:

Bolt size Inches	Torque Inch-pounds
3/8	170
1/2	360
5/8	600

Table IV - Bolting of connection bars - size and number of bolts.

Size bars	Size of bars (inches)						
	3/4	1	1-1/2	2	3	4	5
Inches	Inch	Inch	Inch	Inch	Inch	Inch	Inch
3/4	One 3/8	-------	-------	-------	--------	-------	-------
1	One 3/8	One 1/2	-------	-------	--------	-------	-------
1-1/2	One 3/8	One 1/2	One 5/8	-------	--------	-------	-------
2	Two 3/8	Two 1/2	One 5/8	One 5/8	--------	-------	-------
3	Two 3/8	Two 1/2	One 5/8	One 5/8	Two 5/8	-------	-------
4	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Two 5/8	Four 5/8	-------
5	-------	-------	-------	-------	Three 5/8	Five 5/8	Five 5/8
6	-------	-------	-------	-------	Three 5/8	Five 5/8	Five 5/8

3.17.13 Painting of bus bars. - The bus work shall be painted with insulating material in accordance with 3.26.4.

3.17.14 Spacing of bus supports. - The bus work shall be designed to withstand the maximum root mean square (r.m.s.) asymmetrical short circuit current calculated by the following method:

$$I_{max.} = \frac{4.0 I_n + K_2 E_g}{\sqrt{R^2 + X^2}}$$

Where:

$I_{max.}$ = maximum available asymmetrical r.m.s. short circuit current at one-half cycle, amperes.

I_n = sum of the full connected load currents of the maximum number of induction and synchronous motors assumed to be connected to the system at instant of short circuit, amperes.

K_2 = factors dependent on the ratio of the total reactance X to the total resistance, R, up to the point of fault to be taken from figure 10.

E_g = to rated line-to-neutral voltage of generator r.m.s. volts.

R = total resistance in ohms per phase (parallel and series combinations) up to the fault including the armature resistance of each generator and all significant circuit resistances (cables, circuit breakers, transformers and bus work) between the generator and the fault.

X = total reactance in ohms per phase (parallel and series combination) up to the point of fault using generator direct-axis subtransient reactance of each generator and including all significant circuit reactances (cables, circuit breakers, transformers, and bus work) between the generator and the fault.

For the calculation of required spacing of bus supports to withstand the short circuit current, the value of maximum stress in the outside fibers shall be based on the yield strength of the copper and not upon the tensile strength.

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3.17.15 Bus bar capacity. - Buses shall be designed on a basis of generator capacity and on feeder circuit breaker and switch sizes and capacity. For a single generator, the generator bus shall have a capacity equal to the continuous rating of the generator. For more than one generator, where all generators feed through a common section of the bus, the capacity of the bus for the first generator shall be the same as for a single generator installation; for each subsequent generator, the bus capacity shall be increased by the amount of the added generator's continuous current rating. The capacity of connection buses for each generator unit shall be equal to the continuous rating of the generator. For feeder sections protected by a back-up circuit breaker, the buses shall have a capacity equal to the current rating of the feeder back-up breaker. Otherwise, for feed buses the capacity shall be equal to the combined full load currents of all the feeder circuits, plus the frame sizes of all spare circuit breakers, all multiplied by an approved demand factor based on the types of service of the feeders. An exception to this is that the feeder buses shall not have greater capacity than the generator buses which supply them. Feeder buses shall be tapered wherever practicable. For individual circuits having a type ACB or a 600-ampere frame size, type AQB circuit breaker, the bus connections shall have a capacity based on the circuit breaker. For individual circuits having a 100-ampere or a 250-ampere frame size, type AQB circuit breaker, or having knife switches, the bus connections shall have a capacity based on the frame size of the circuit breaker or on the ampere rating of the switch.

3.17.16 Cable connector types. - Cable connectors installed on all cables leaving and entering the switchboard shall be of the solderless type, in accordance with Specification MIL-E-16366. For all cables of 75,850 cm. and smaller, the connectors may be of a type which clamps the cable at right angles to the direction of the circuit breaker or switch stud (see figure 9). Cable connectors shall not be installed on spare circuit breakers to which a cable is not attached. For bus connections within the switchboard, using cable solderless type connectors satisfactory to the bureau or agency concerned shall be used so far as is practicable. For all solder joints made during construction of the switchboard, solder in accordance with Specification QQ-S-571, shall be employed. For all cables of 4000 cm. and larger, cable connectors shall be secured by means of locking devices in accordance with 3.12.12.3.

3.17.17 Connections between buses. - Although the switchboard (switchgear group) is constructed in sections, the switchboard main bus is a single circuit with portions on each section. Connections between the buses of adjacent sections of a switchgear group are made by cables; instead of solid bus bars, to permit relative movement of the sections under shock without rupturing the switchboard main bus circuit. A means for readily disconnecting the cables from the bus shall be provided in all sections having a source of power other than from the connecting cables. The cable connections shall be made by the installing activity. Because of the vital nature of the cables interconnecting switchgear group sections, and the variety of machinery arrangements encountered, careful consideration shall be given the special design arrangements for these connecting cables. The installation shall conform to the following:

(a) The switchboard design arrangement shall include means to provide for the entrance of the connecting cables into the switchboard structure, and for connecting the cables to the switchboard bus. This connection shall be made through a disconnecting device where the section has a source of power other than from the connecting cables.
(b) The connecting cables shall have a current rating equal to that of the smaller of the two connected buses. Three conductor, heat and flame resistant (type SGA of Specification MIL-C-2194) connecting cables shall be used. Where several cables in parallel are required, the three phases shall be carried in each cable. For d.c. vessels, one, two, or three-conductor, type TSGA cable, as applicable, shall be used.

3.17.17.1 Consideration shall be given to facilitate connection (by the installating activity) of the ship's cables to the switchboard so that these cables will not result in restriction of access features. Armor will be removed from the portion of ship's cables inside the switchboard. Instructions for interconnecting wiring shall be shown on the switchboard drawings.

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3.17.18 Cables entering switchboards. - Cables entering the switchboard through the top or bottom drip shields will be provided with standard terminal tubes. In general, cables entering the switchboard from the top should enter at the rear, beneath the top drip shield, and a slack cable loop provided in each cable to insure that moisture collecting on the cable will not drain into the switchboard. The switchboard design arrangement shall provide for proper cable connections without restriction of access features.

3.18 Wiring of control and instrument circuits. -

3.18.1 Control and instrument cable types. - The cables used for control and instrument circuits, ground detector and indicator lights and other parts, shall conform to type SHFS of Specification MIL-C-915. Type SHFS-3(19) shall be used where the rated current carrying capacity is adequate for the circuit application. The extra flexible (SHFS-3(19)) wire shall be used for all wiring between hinged panels and the stationary structure. The circuits shall be color coded with solid color braid as follows:

(a) Current transformer, secondary circuits ---------- Red.
(b) All 450 volt circuits ---------- Black.
(c) All other circuits, control, potential, indicator lights, etc. ---------- Gray.

3.18.2 Cable end connectors. - The ends of each wire shall be connected to the apparatus studs by means of approved type solderless terminals or by forming a wire loop around the apparatus stud or terminal screw and retaining the loop in a cup or crimped washer. The wire loop shall be soldered to secure the strands together. Locking devices in accordance with 3.12.12, shall be used to secure the nuts or screws connecting wire loops to the apparatus connection point. Refer to figure 11 for illustration of wire connections.

3.18.3 Forming and securing cable groups. - The wires shall be neatly formed into groups which shall be laced or corded and supported or clamped in a manner which will prevent chafing of the insulation due to vibration. Commercial nylon cable straps or metal clamps may be used, but where metal is used it shall be covered with a noninflammable material which will prevent chafing of the wire insulation. Lacing material shall be either nylon straps, noninflammable cord, glass fiber tape type SR-1, Specification MIL-I-3158, or pressure sensitive adhesive plastic tape, Specification MIL-I-7798. The last two materials are specifically adapted for serving switchgear control wiring in locations subject to abrasion. Glass fiber tape may be coated with adhesive on one side, but when used with this adhesive coating, it shall be covered with pressure sensitive adhesive plastic tape for permanent adhesion. There shall be no splices in wire and all connections shall be made at the terminal studs of the devices or terminal boards. The wire groups running from hinged panels shall be formed and clamped so that sharp bends do not occur with the panel in either the open or closed position. Where wires run from one section of a switchboard to an adjacent one, and these sections are separated for shipping or installation purposes, terminal boards shall be provided on one of the units and located so as to facilitate completion of the interconnections during installation of the switchboard on the vessel.

3.18.4 External control cable connections. - External control cable connections to the switchboards shall terminate at terminal boards accessibly located near the top or bottom of the switchboard, corresponding to ship's cable entrance. Where calibrated shunt leads are required, they shall terminate on the instrument studs. An excess length of shunt leads shall be neatly coiled and secured so as not to interfere with access to other equipment. Circuitous routing of control and instrument wiring shall be avoided. Small wiring to circuit breaker trip coils shall run direct to the coils and associated contact circuits and shall not be routed through terminals of other apparatus. Taps from bus bars shall be made by bolting terminals to the bus bars.

3.18.5 Synchronizing control circuits. - Where synchronizing facilities are required, the closing control circuit of electrically operated circuit breakers (a. c. generators and bus ties) shall be connected through contacts of the corresponding synchronizing switch or switch position so that the proper synchronizing potentials are connected through the switch before the breaker closing circuit is completed, thus guarding against inadvertent closing of the breaker. For increased reliability of synchronizing-

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facilities and standardization of practice, the synchroscope shall be supplied by the phase A-B transformers, and the synchronizing lights by the phase B-C transformers provided for metering, synchronizing, and relaying potential as shown on figure 12. The loss of one transformer shall not disrupt both synchronizing facilities.

3.18.6 Space heater circuits.- When space heaters are provided with the associated generator, the heater circuit shall be supplied through a heater control switch from a local emergency lighting circuit. An indicator light (white) shall be provided adjacent to the switch to indicate when generator heaters are energized.

3.18.7 On surface ships, the indicator lights of the mimic bus shall be supplied from a local lighting system which has available both normal and emergency power. On submarines, power for the indicator lights shall be taken from the a.c. or d.c. control power buses.

3.18.8 The control transformers shall be supplied in number as required for supplying governor motor control circuits.

3.18.9 Ground detector lights shall be connected as shown on figure 13.

3.18.10 The voltage regulator equipment shall be connected in accordance with the master drawings approved by the bureau or agency concerned.

3.18.11 Wire and terminal marking.- Internal wiring of equipment shall be marked by stamping on wire terminals or, where terminals are too small to permit stamping, wire markers of synthetic resin tubing, type F, grade A, Specification MIL-I-631 or fiber tags are satisfactory. Markings shall agree with the designation shown on the approved switchboard drawings to facilitate checking of connections. It will not be necessary to mark the terminal studs on instruments, and control switches, but wiring diagrams shall clearly indicate the studs or contacts to which marked wires are connected. Terminal blocks shall be equipped with marking strips. These marking strips shall be permanently marked at each terminal to agree with the wire marker number on the wire to that terminal and also to agree with the corresponding number on the wiring diagram.

3.18.12 Shunt trip coil circuits (both a.c. and d.c.) for tripping type ACB circuit breakers shall be routed through a normally open contact (open when the breaker is open) of the circuit breaker auxiliary switch, so that the trip coil cannot remain energized after the circuit breaker is open.

3.19 Circuit breakers.- Circuit breakers shall conform to Specifications MIL-C-17587 (for type ACB), MIL-C-17361 (for type AQB), and MIL-C-17588 (for type ALB).

3.19.1 Mounting (AQB and NQB).- The types AQB and NQB circuit breakers for switchboard use shall be equipped with removable type contacts on both line and load side, connections being made by use of terminal mounting blocks in accordance with Specification MIL-C-17361. The switchboard mounted circuit breakers shall be vertically mounted with the line side terminals at the top. Horizontal mounting is permissible only on cabinet type power and lighting panels. Types AQB and NQB circuit breakers on switchboards and load center units shall be mounted on the minimum center spacing as shown on figure 1. When interlocks are required for types AQB and NQB circuit breakers, they shall be front of panel mounted and shall be suitable for use with circuit breakers of any make by no further modification than possible reassembly of the interlock.

3.19.2 Mounting (ACB).- Type ACB circuit breakers shall be drawout mounted, and shall be fitted with the necessary contact, mounting, and positioning equipment. The drawout mechanism shall be so designed to provide for drawout operation of any type ACB circuit breaker of a given frame size. Steel drawout mechanism may be used with aluminum constructed switchgear.

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3.19.3 Drawout mechanism. - Each mechanism shall allow an operator to move the circuit breaker horizontally without lifting the circuit breaker or contacting any live parts. The positions for the removable carriage shall be provided as follows:

(a) "Connected" position in which the power circuit contacts and the control circuit contacts are engaged and breaker is ready for operation. In this position, the carriage shall be securely clamped to the frame of the fixed structure to prevent relative movement which might cause damage under conditions of high impact shock.
(b) "Disconnected" position in which the carriage will extend from the compartment sufficiently to allow inspection of the disconnecting devices on the carriage and so that the circuit breaker may be lifted from its supports and lowered to the deck.

Positive means shall be provided for securing the carriage in each position and in all intermediate positions, so that should the operator release the carriage, it will automatically be locked in that position and its position not be changed by the roll of the vessel. Positive means shall also be provided to prevent the carriage from being withdrawn from the "connected" position, or replaced beyond a 1-inch air insulation distance between the drawout contacts, except when the circuit breaker is in the open position or except when the locking device is manually disengaged or removed by the operator. Neither shall it be possible to close the circuit breaker when the carriage is in such a position that the drawout contacts have less than 1-inch air insulation distance, until the carriage reaches the "connected" position. The drawout mechanism shall be constructed in accordance with Drawing 9000-S6205-74338, or shall be of a type specifically approved by the bureau or agency concerned. In case of drawout mechanisms for 600 and 1600 ampere frame size circuit breakers, the mechanism design shall be such as to mount any approved make of circuit breaker.

3.20 Switches, disconnecting devices, and transfer switches. -

3.20.1 All switches (knife, tumbler, control, instrument) shall be in accordance with Specification MIL-S-18396 or MIL-S-15291. All knife switches shall be equipped with a locking device as shown on Drawing 9000-S6202-73063, to hold them in the open and closed positions against shock.

3.20.2 All instrument and control switches shall have gray or natural light tan escutcheon plates with black lettering. The switches shall be provided with five styles of handles to identify the function for which the switch is provided. The handle styles and corresponding function shall be as follows:

Radial - governor control.[1]
Oval - synchronizing and voltage regulator.
Round (plain) - rheostat control.[1]
Round (knurled) - instruments (voltmeter, ammeter, frequency meter, temperature indicator); test [1](emergency switchboard).
Pistol-grip control circuit breaker[1], ground lamps[1], control bus selector, lighting and ventilation cutout, and motor restart.[1]

[1]Shall have spring return handles.

3.20.3 All disconnecting devices for isolating generator and bus-tie circuit breakers and for performing other disconnecting functions shall be of the type in which contact surfaces are forced together by pressure applied by means of screw threads. With full rated current through the disconnect devices, the maximum temperature rise of the devices shall not exceed 30° C. The disconnect devices shall be mounted inside the switchboard and shall be readily accessible after removal of the rear enclosure. Tools (if required) for operation of the devices shall be mounted outside of the switchboard enclosure adjacent to the panel that must be removed for operation of the device.

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3.21 Fuses and terminal blocks.-

3.21.1 Fuses shall be in accordance with Specification MIL-F-15160. Fuses rated 250 V. may be used for all applications where the available short circuit current does not exceed its safe interrupting capacity. Identification plates for silver sand fuses shall state that these fuses shall be replaced only with silver sand fuses, or a warning plate clearly pointing this out shall be added.

3.21.2 Fusing of potential transformers.- Each metering potential transformer shall be fused on the primary side with a separate 6 AMP fuse in each leg of each transformer, so that a fault on one transformer will not affect the other. Potential transformer secondary circuits shall not be fused, except where a circuit leaves the switchboard for remote metering. In such a case, the ungrounded portion of the circuit which leaves the switchboard shall be fused, and the fuses shall be located so as not to open any circuits on the local switchboard. One 6 ampere fuse shall be provided in each leg of each indicator light transformer connected to a main bus or bus tie for isolation of the transformer, should the primary winding or connections become short circuited. The control circuits for electric operation of circuit breakers shall be fused for short circuit protection only.

3.21.3 Fuse mounting.- The fuses mounted on switchboards shall be accessibly located for replacement (see figure 14). The fuses shall be grouped on subpanels insofar as practicable, but the length of leads between fuses and the power circuit taps shall be kept to a minimum. The fuses for 450 V. circuits shall be enclosed by hinged steel covers. Means shall be provided to secure fuses against dislodgment from the fuse clips due to shock. Where hinged steel covers with insulating strips for restraining fuses are not provided, the fuse retainer, type B, shown on Drawing 9-S-5312-L, shall be used. Identification plates shall be provided for each set of fuses, and shall indicate the circuit controlled, the phases or polarities, and the fuse ampere rating.

3.21.4 Fuse holders.- The clip type fuse holders (other than for live-front switchboards) shall be enclosed by hinged steel covers. The use of enclosed fuse panels is also preferable for fuses used in control, metering and indicating lamp circuits rated not over 120 V; however, in those cases where the use of fuse boxes causes inadequate access to the unit for inspection and maintenance, or requires long lengths of wire for making such connections, dead-front fuse holders shall be used in lieu of the fuse boxes. These fuse holders may also be used for similar applications on 120 and 240 V. d.c. circuits, where the available short circuit current does not exceed safe interrupting capacity of the fuse. The fuse holders shall conform to types 10FH1, 10FH2 and 12FH1 of Drawings 9000-S6202-74228, 9000-S6202-74229 and 9000-S6202-74230. Dead-front fuse holders shall not be installed for any distribution circuit fuse on power or lighting distribution switchboards or panels.

3.21.5 Terminal boards.- To minimize the possibility of breakage of the terminal boards, due to distortion of the mounting support under shock, the length of the terminal board shall not exceed 12 terminal points.

3.22 Instruments, relays and instrument transformers.-

3.22.1 Electrical measuring instruments.- Electrical measuring instruments shall conform to Specifications MIL-I-1361, MIL-T-2867, MIL-M-16034, MIL-M-16125 and MIL-W-19088, having a 4-1/2 inch, rectangular light gray colored case to approximate the color on the switchboard panel, with 250-degree scale. Instrument scales shall be white with black markings and lettering, and shall be marked to indicate the full load and normal voltage calibration point as required by Drawing 9-S-4907-L. Figure 15 shows typical mounting of instruments.

3.22.2 Instrument transformers.-

3.22.2.1 Current transformers for metering and relaying shall conform to Specification MIL-I-1361, having an accuracy for Navy secondary standard N.S.S. rating. Potential transformers shall have a minimum rating of 100 V. AMP based on accuracy and 300 V. AMP based on thermal capacity. Autotransformers or other transformer type devices used to provide phase angle changes for single phase watt-meters on three-phase circuits or similar functions shall conform to Specification MIL-I-1361.

3.22.2.2 Transformers shall be provided for supplying metering, synchronizing, and relaying.

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3.22.2.3 Instrument transformers provided for voltage regulators shall not be used to supply any other burden, unless specifically approved by the bureau or agency concerned. For each group of instruments, relays, and synchroscope, one set of instrument transformers shall be used unless the burden on one set would be too great for the accuracy required. The potential transformers for each group of instruments, relays, and synchroscope shall not be used for other potential devices without approval by the bureau or agency concerned. Where two potential transformers are used for three phase operation, they shall be connected open delta on the primary and secondary. One transformer shall be connected across phase A-B and the other across phase B-C.

3.22.2.4 Where current transformers are required to supply 3-phase current, two transformers shall be used, one connected in phase A, one in phase C. Their secondaries shall be connected open delta. Current transformer secondary circuits which extend beyond the section containing the current transformers shall be provided with protective devices conforming to Specification MIL-I-1361, to prevent high voltage in the event of an open circuit. The protective devices shall be connected so that their operation will not shunt out protective relays which may be connected in the circuit. A short circuiting switch shall be connected in parallel with the protective device for manually short circuiting the remote part of the current transformer circuit. These connections are shown on figure 16.

3.22.2.5 One secondary terminal of each metering and relaying potential, current, and indicator light transformer shall be grounded, except for 3-phase connections, where one ground connection shall be made to the common, or B phase of the two transformers.

3.22.3 All relays, both protective and auxiliary types, shall conform to Specification MIL-R-2033 and shall be of types which have been tested and approved for class HI shock resistance. Relays shall have the following protective functions when used with a suitable circuit interrupting device:

Overpower protection.
Reverse power protection.
Long time delay overcurrent protection.
Current phase balance protection.
Undervoltage protection.
Ground fault protection.

3.23 Illumination and indicator lights. -

3.23.1 In general, proper switchboard illumination will be obtained by general room illumination.

3.23.2 Indicator lights shall conform to Drawings 9000-S6202-73907 for a.c. and 9000-S6202-74307 for d.c.

3.23.3 For position indication of circuit breakers, blue indicator lights shall be used to show that the circuit breaker is closed. All other indicator lamps shall have color caps as specified for the particular application. The color code shall be as specified in table V.

Table V - Globe color designations.

Red	Danger or emergency
Green	Normal
Clear, etched (white)	Power on or power available
Blue	Closed
Clear (not etched)	Synchronizing and ground detector lights
Amber	Abnormal
Yellow	Open

3.23.4 Resistors for use with indicator lights, ground detector lights and other functions shall be in accordance with Specification MIL-R-26. All indicator lights shall be furnished complete with lamps.

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3.24 Switchboard enclosures.-

3.24.1 Rear enclosures.- The rear of each unit shall, when so specified, be provided with a removable enclosure for protection of personnel and to prevent contact with the live parts. These rear enclosures shall be made of expanded metal (rolled flat) or perforated sheet, and sectionalized into not less than two horizontal parts to facilitate handling. The enclosures shall be adequately secured to prevent rattle but shall not be used for structural strength. The openings in expanded metal or perforated sheet shall not be used for structural strength. The openings in expanded metal or perforated sheet shall not pass a rod having a diameter greater than 3/8 inch. Provision shall be made for removal of fuses without removing the rear enclosure of a unit. The removable rear enclosures as required for the rear of each unit may be omitted when specified (see 6.1). On bench type control units having equipment mounted on the rear enclosures, the rear enclosures shall be solid sheet. Clearances between live bus work and disconnect devices, or both, and rear enclosures shall be adequate, in order to prevent these nonstructural strength panels from coming into contact with live bus work, if the panels are buckled inward.

3.24.2 Top enclosure.- The top of each section shall be enclosed by a sheet cover. The top shall be of suitable design to prevent dripping water or falling objects from entering the switchboard enclosure or damaging instruments on the front panels. It shall extend approximately 4 inches out over the face of the front panels and 4 inches beyond the rear of the switchboard structure. The clearance between the upper edge of the front panels and the under side of the top sheet, where it extends over the front, shall be at least one-half inch. The front and rear edges of the top sheet shall be turned up to form a 1-inch flange along the entire length of the section or switchboard to prevent water from spilling over the front or rear. The end edges of the top sheet shall be turned down to form a flange, shall overlap the side sheets of the switchboard to allow drainage over the end, and to prevent water from seeping down the inside of the side sheet. To insure watertight integrity at the corners, the turned-up front and rear flanges shall be extended around the corners and along the end of the top sheet to overlap the turned-down flange by approximately 4 inches. The joint formed between the bottom of the extended flange and the end of the top sheet shall be welded to form a watertight seam. A typical drip-shield is shown on figure 17.

3.24.3 Panel fastening.- Hinged panels on the front and rear of the unit shall be secured by thumb screws made captive in the panel (see typical figures 18 and 19), or by fasteners in accordance with Specification MIL-F-5591. Thumb screws shall be of a type which can be quickly removed without use of tools, to gain rapid access to the interior of the unit. Removable panels on the front and rear of the unit shall be secured by hex head bolts or thumb screws made captive in the panel.

3.25 Guard rails.-

3.25.1 Each switchboard unit, except for distribution units, shall be provided with a vertical grab rod located on the right hand side, facing the unit, so that an operator will have means of support while operating the switchboard (see figure 20). The rod shall be of smooth hardwood, supported with a sufficient number of brackets to provide adequate support for personnel under all operating conditions of the vessel. The rod shall be attached in such a manner that all panels can be removed and hinged panels opened, without removal of the rod from the member to which it is attached. These rods shall be located so that they will not interfere with operation of the equipment and shall not extend more than 9 inches from the front of the panel.

3.25.2 Removable guard rail sections shall be provided on the rear framework of the switchboard (see figure 21). In general, two horizontal rails will be required at proper levels to prevent personnel from being thrown against live buses, due to the motion of the vessel. They shall be mounted on the inside of the rear vertical members so as not to protrude beyond the rear or ends of the structure. These guard rail sections shall be easily removable for maintenance access but shall be provided with securing bolts or other locking means to prevent dislodgment due to shock.

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3.26 Painting. -

3.26.1 General. - The exterior and interior surfaces of all enclosures shall be painted. The order of operations shall be as follows:

(a) Complete all fabricating operations, such as welding, machining, drilling, and tapping.
(b) Remove all rust and other visible corrosion products.
(c) Remove all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.
(d) Apply primer pretreatment coating or chemical treatment and primer.
(e) Apply enamel.

3.26.2 Primer. - The primer shall be applied in accordance with either of the following two methods:

(a) One coat of primer pretreatment coating in accordance with Specification MIL-C-15328, applied as a continuous film 0.0002 to 0.0005 inch thick.
(b) A phosphate treatment conforming with grade I of Specification MIL-C-490, followed by one coat of primer in accordance with Specification TT-P-664, JAN-P-735 or MIL-P-8585, applied as a continuous film 0.0004 to 0.001 inch thick.

3.26.3 Enamel. - Two coats of gray enamel conforming with class 2, type II or III, (with Specification JAN-P-735 primer use type I) of Specification MIL-E-15090 shall be applied as a continuous film, each approximately 0.001 inch thick.

3.26.4 Bus bars. - Copper and aluminum bus bars on switchboard shall have two coats of red colored insulating paint, conforming to Specification MIL-I-17384, or other specifically approved paint. Coating shall be flexible, tough, moisture and acid resisting, and shall have an insulation resistance value of at least 500 volts per mil of thickness. Two coats, each 0.004 to 0.006 inch thick, shall be applied. Before painting, burrs and sharp edges shall be removed and bus bars cleaned to remove dust, grease, oil, or other foreign material. Paint may be applied by spraying or may be flowed on with a soft bristle brush when spraying is impracticable. Coatings shall be free from pinholes. Bus bars shall be painted before assembly up to one-quarter to one-half inch from the joint area. Where a painted area of a bus bar would be an inch or less in width, it need not be painted. Do not paint bus joints or connectors either before or after assembly. Taping of bus bars located entirely within the switchboard enclosure is prohibited.

3.27 Identification plates and marking. -

3.27.1 Identification plates shall conform to types A, B, C, F or H of Specification MIL-P-15024. Plastic identification plates shall be light gray. Inscription or markings for all type plates shall be black.

3.27.2 Location. - The identification plate shall be placed in close proximity to the operating handle of the equipment to which it refers, and generally either directly above or directly below it.

3.27.3 In general, identification plates shall be of a size as listed in Specification MIL-P-15024. Other sizes of plates may be used when specifically approved.

3.27.3.1 Thickness of identification plates shall be in accordance with Specification MIL-P-15024.

3.27.3.2 Identification plates for rheostat handwheels and other rotary devices shall, where required, be circular or other shape to suit the equipment.

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3.27.4 Identification plates employing lettering of size and giving information consistent with the following shall be provided at or near the top on the front enclosure of each switchgear section. The lettering shall be in Gothic type capitals:

Name or functional designation of item; letter size ratio of 3:1.
Manufacturer's name; letter size ratio 1:3.
Manufacturer's drawing number (front view); letter size ratio 1:3.
Year manufactured; letter size ratio 1:3.
Bureau or agency drawing number (front view); letter size ratio 1:3.
Bureau or agency stock number; letter size ratio 2:3.
Circuit identification plate letter size: 1-1/2 by 5 inch identification plate.
Apparatus or circuits controlled, etc., 1/8 inch letters.
Feeder letter and number, for example; F-438; 3/8 inch letters.
Identification of source, for example,
Fed from power panel 3-69-3; 1/8 inch letters.
1 inch by 3 inches identification plate:
First line circuit number for example,
1-FB-401; 1/8 inch letters.

3.27.5 Unless lack of space prevents, abbreviations of words on identification plates shall be avoided except in case of words which have abbreviations in common use in accordance with Standard MIL-STD-12.

3.27.6 Installation.- Identification plates shall be fastened securely to such parts of the equipment as ordinarily will not be renewed during its service life. Identification plates shall be attached in accordance with Specification MIL-P-15024.

3.27.7 Detail application requirements.- An identification plate shall be provided for each complete switchboard section, mounted on the front of the switchboard near the top center. Identification plates shall be installed immediately above each feeder switch and circuit breaker and shall contain the following information:

Circuit name.
Feeder designation and number.
Feeder load, amperes.
Circuit breaker element rating or fuse size.

3.27.8 Identification plates without markings shall be provided for mounted spare feeder switches and circuit breakers.

3.27.9 Instruments shall be marked with the name of circuits; marking may appear on instrument or on a separately mounted identification plate.

3.27.10 Fuses for each switchboard control circuit shall have an identification plate showing name or designation of circuit, and fuse size. Fuses for instruments and instrument transformers shall have plates showing instrument or circuit designation and fuse size in amperes.

3.27.11 Marking switchboard buses and terminals.- Buses shall be marked in a visible location with their polarity or phase identification.

3.27.12 Terminals on the back of each switchboard shall be marked with the circuit letter and number and with polarity or phase identification by identification plates adjacent to the terminals. Where the size of such plates or their fastening to surfaces adjacent to terminals makes their use impracticable, the required marking shall be marked directly on the back of the switchboard in a visible location adjacent to terminals, or on their bus connections.

3.27.13 Positive, neutral, and negative polarities of direct current buses and terminals shall be indicated by +, ±, and - signs, respectively.

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3.27.14 Phase identification of alternating current buses and terminals shall be indicated by the capital letters A, B or C, as applicable. A neutral bus or terminal of a four-wire, three phase circuit, where used, shall be indicated by the capital letter N. The letters A, B and C, in that order, shall indicate the phase sequence.

3.28 Drawings.- Drawings shall be type I or type II as specified in Specification MIL-D-963.

3.28.1 Type I - master drawings.- Master drawings are complete drawings of standardized switchgear units conforming to Drawing 9000-S6201-74453. They shall be submitted by the manufacturer in accordance with type I of Specification MIL-D-963, and shall cover the following:

(a) Dimensional front plan, side section and rear views of the switchboard unit, showing standardized overall dimensions of the unit, number, size and location of holes for interframe bolting, location of required equipment, location and size of main busses passing through the unit, and connections thereto. These views, together with the list of material, shall show the location and function of all appurtenances comprising the unit. The front and plan view of the unit shall, in addition, show all limiting dimensions including clearance necessary for hinged panels and all apparatus mounted on the front.
(b) A dimensional rear elevation view of the unit showing casualty power equipment and rear enclosures where specified.
(c) A dimensional floor or mounting view to show arrangement of anchor bolt drilling.
(d) Supplementary views as required show the necessary dimensions and equipment.
(e) Identification plate list of parts having general application; complete list of material with name of all parts, part numbers, bureau or agency certificate of approval or drawing numbers of specification numbers, together with material for all parts included in standardized units. Total weight of each unit, with and without circuit breakers, shall be shown on drawing. All switches, circuit breakers, contactors, and resistors, shall be marked to show their current carrying capacity and voltage rating, where applicable.
(f) A one-line wiring diagram of main power circuits. Circuit breakers, disconnect links, main switches or other power circuit switching equipment shall be shown as required. Main and interconnecting busses may be shown.
(g) Detail drawings of appurtenances as required shall be shown or referenced. Where appurtenances are required to be of approved types for which test record drawings or certification data sheets are furnished the bureau or agency concerned, no working or corrected drawings of such appurtenances will be required; however, the shipbuilder or contractor shall reference each test record drawing or certification data sheet covering appurtenances, and the items in question shall conform completely to such drawings or certification data sheets, unless otherwise specified. If supplementary assembly, outline, or detail drawings of such items are necessary, in the judgement of the bureau or agency concerned, they shall be furnished by the switchboard manufacturer.
(h) Typical construction methods of the switchboard unit, showing details of framework, details of attaching front panels, details of joining structural members, location and types of welding, and details of drawout mechanisms for circuit breakers shall be shown or referenced.
(i) The drawings shall show piece numbers, identification plate numbers, and device numbers where necessary, and these shall agree with the corresponding numbers on the identification plate list and list of material. The list of material shall show the manufacturer's name and all sufficient information to identify readily the equipment in case replacement or purchase of new material is necessary.

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3.28.2 Type II - construction drawings (ship's drawings). - Construction drawings shall be submitted by the manufacturer in accordance with type II of Specification MIL-D-963 and shall cover the following:

3.28.2.1 Ship's drawings - switchboards using standardized units. -

(a) A dimensional front and plan view of the switchboard, showing assembly of all units comprising the switchboard, and overall dimensions of each section type numbers of units as shown on Drawing 9000-S6201-74453 shall also be shown.
(b) Views as required showing necessary details not shown in the master drawings of the units, such as interconnection of busses between unit, dimensions and spacing of footing or mounting members, and other details.
(c) Identification plate list of parts having specific application; list of material for all specific and interconnecting parts not covered by master drawings, showing all required additional information, such as fuse or trip element data.
(d) Complete wiring diagram, showing bus sizes, feeder numbers, terminal blocks, circuit breaker, switch and fuse sizes, cable sizes and nameplate numbers. All electrical appurtenances shall be marked or indicated by piece numbers, to identify them readily with the corresponding part on the list of material. Where five or more single lines used to represent several leads parallel each other on the plan, they shall be separated into groups (two or three to a group as convenient), to facilitate following with the eye, instead of having them all uniformly spaced.
(e) A single line diagram of metering, relaying, and control circuits. This diagram shall show the major bus circuits, the number and location of all instrument and control transformers, the instruments, relays and other equipment connected to each transformer, and the location of the instruments in relation to the various switchboards. The diagram shall be included among the drawings in the technical manual.
(f) Detail drawings of appurtenances not covered by master drawings, as required. Where appurtenances are required to be of approved types for which test record drawings or certification data sheets are furnished the bureau or agency concerned, no working or corrected drawings of such appurtenances will be required; however, the switchboard manufacturer shall reference each test record drawing or certification data sheet covering appurtenances, and the items in question shall conform completely to such drawings or certification data sheets, unless otherwise noted. If supplementary assembly, outline, or detail drawings of such items are necessary, in the judgement of the Supervisor of Shipbuilding for the purpose of proper inspection or the identification of repair parts, they shall be furnished by the switchboard manufacturer.
(g) A typical construction drawing showing necessary details not covered by 3.28.1 (h) or by other drawings.
(h) The drawings shall show piece numbers, identification plate numbers, and device numbers where necessary, and these shall agree with the corresponding numbers on the identification plate list and list of material. The list of material shall show the manufacturer's name and all sufficient information to identify readily the equipment in case replacement or purchase of new material is necessary.

3.28.2.2 Ship's drawings - switchboard using nonstandardized units. -

(a) A dimensional front and plan view of the switchboard, showing all limiting dimensions including clearance necessary for hinged panels and all apparatus mounted on the front.
(b) A dimensional rear and plan view of the switchboard, showing bus structure, bus sizes and spacing, supports, and assembly methods.

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(c) Dimensional section views, front, rear, and side vertical; and horizontal section views; in sufficient number to show arrangement and location of equipment and busses; clearances; and sizes of steel frame members. Proposed arrangements for incoming cables and feeder cables shall be shown with supports and methods of installation.

(d) A dimensional floor or mounting view to show arrangement of footing or mounting members and anchor bolt drilling.

(e) Identification plate list with fuse or trip element data; complete list of material with names of all parts, part numbers, bureau or agency certificate of approval or drawing number or specification number, together with material and total weights. All switches, circuit breakers, contactors, and resistors, shall be marked to show their current-carrying capacity and voltage rating, where applicable.

(f) Complete wiring diagram, showing bus sizes, feeder numbers, terminal blocks, circuit breaker, switch and fuse sizes, cable sizes and nameplate numbers. All electrical appurtenances shall be marked or indicated by piece numbers, to identify them readily with the corresponding part on the list of material. Where five or more single lines used to represent several leads parallel each other on the plan, they shall be separated into groups (two or three to a group as convenient), to facilitate following with the eye, instead of having them all uniformly spaced.

(g) A single line diagram of metering, relaying, and control circuits. This diagram shall show the major bus circuits, the number and location of all instrument and control transformers, the instruments, relays and other equipment connected to each transformer, and the location of the instruments in relation to the various switchboards. The diagram shall be included among the plans in the technical manual.

(h) Detail drawings of appurtenances as required. Where appurtenances are required to be of approved types for which "Test Record Drawings" are furnished the bureau or agency concerned, no working or corrected drawings of such appurtenances will be required; however, the shipbuilder or contractor shall reference each "Test Record Drawing" covering appurtenances, and the items in question shall conform completely to such drawings, unless otherwise specified. If supplementary assembly, outline, or detail drawings of such items are necessary, in the judgment of the Supervisor of Shipbuilding, for the purpose of proper inspection or the identification of spare parts, they shall be furnished by the shipbuilder or contractor.

(i) A typical construction drawing, showing details of framework, details of attaching front panels, details of joining structural members including bolting, location and types of welding, details of fuse boxes, number and size of bolts used for supporting breaker bases, breaker mounting straps and bus supports, details of hinging panels including latches and stops, methods of supporting copper, and type of guard rails and other details of construction not shown in sufficient detail on other drawings.

(j) The drawings shall show piece numbers, identification plate numbers, and device numbers where necessary, and these shall agree with the corresponding numbers on the identification plate list and list of material. The list of material shall show the manufacturer's name and all sufficient information to identify the equipment readily in case replacement or purchase of new material is necessary.

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3.29 Technical manuals.- The technical manuals shall conform to type II of Specification MIL-M-15071 except as follows:

(a) Cover.- The cover shall include the class of vessel and each type of switchboard. The size of print for the equipment shall be 14 points, except the word "switchboards", which shall be 30 points.

(b) Instruction leaflets of switchboard mounted devices for inclusion in the switchboard technical manuals shall be arranged in individual alphabetical sections describing the functions of the devices. Only relevant matter shall be included. The manufacturer's trademark shall be inscribed in the upper right hand corner of each page of the leaflet to permit ready identification of the correct instructions where similar devices of different manufacture are installed on the same switchboard.

(c) Switchboard technical manuals shall, unless otherwise specified in the contract or order, contain all the information required in the following paragraphs and in the order listed: (Note.- The above applies where instructions for several switchboards are included in one book. If any of these switchboards are not furnished, the following chapter numbers shall be rearranged accordingly.)

Part 1 - Switchboards:
Chapter 1 - General information.
Chapter 2 - Switchgear group ship's service power.
Chapter 3 - Switchgear group - emergency power.
Chapter 4 - Battery charging switchboard.
Chapter 5 - Steering power transfer switchboard.
Chapter 6 - Miscellaneous. (Other switchboard sections if supplied.)
Part 2 - Air circuit breakers.
Part 3 - Miscellaneous switchboard devices.
Part 4 - Repair parts and tools.

(d) Index.- The index shall be included at the beginning of each part, and shall be printed on a heavy paper spacer. The table of contents in the front of the book shall refer to this index.

Note.- Where list of illustrations and index are included in the device sections prepared by the manufacturer, reference shall be made to the list and index.

3.29.1 Part 1 - switchboards.- The chapter numbers below apply where instructions for several switchboards are included in one book.

Chapter 1 - General information.

(a) Introduction.- The introduction shall include a list of the various switchboards furnished, and description of their locations and construction.

(b) Detailed description.- The detailed description shall include important mechanical details and features which have a bearing on the operation of the equipment.

(c) Principle of operation.- Discussion of principles of operation shall be based upon a simplified one-line diagram of the ship's service system, emergency system, and other systems, and shall include a description of the sources of power, electrical arrangement of buses, purpose of the various switchboards, that is, battle power, general power, lighting, emergency power, and others. This chapter shall include a discussion of protective device arrangement and operation, selective tripping, such general features as synchronizing, ground detection, and indicating lamps, and such illustrations and sketches as are necessary to clarify the operation.

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Chapter 2 - Switchgear group - Ship's service power - operating instructions. Include step by step procedure for starting generators, synchronizing and paralleling generators, dividing loads, securing generators, and use of ground lamps and indicating lamps.
Chapter 3 - Switchgear group - emergency power - operating instructions.
Chapter 4 - Steering group transfer switchboard operating instructions.
Chapter 5 - Battery charging switchboards - operating instructions.
Chapter 6 - Miscellaneous - operating instructions for other switchboard sections, if supplied.

3.29.2 Part 2 - air circuit breakers. - This part shall consist of a chapter for each frame size circuit breaker used on the switchboard, arranged as follows:

Chapter 1 - Type ACB - 250 ---------- 250 ampere frame size.
Chapter 2 - Type ACB - 640 ---------- 640 ampere frame size.
Chapter 3 - Type ACB - 1600 and ACB 1600H ---------- 1600 ampere frame size.
Chapter 4 - Type ACB - 3200H ---------- 3200 ampere frame size.
Chapter 5 - Type ACB - 4000H ---------- 4000 ampere frame size.
Chapter 6 - Type AQB - A250 and LF250 ---------- 250 ampere frame size.
Chapter 7 - Type AQB - A100 and LF100 ---------- 100 ampere frame size.
Chapter 8 - Type AQB - A600 ---------- 600 ampere frame size.
Chapter 9 - Type AQB - A400 ---------- 400 ampere frame size.
Chapter 10 - Type ALB - 5(3 pole) ---------- 50 ampere frame size
Chapter 11 - Type ALB - 10(3 pole) ---------- 200 ampere frame size.
Chapter 12 - Type ALB - 1(single pole) ---------- 50 ampere frame size.
Chapter 13 - Type AQB - A200 ---------- 200 ampere frame size.

Each chapter shall be arranged in individual alphabetical sections describing the functions of the circuit breakers. These instructions shall be prepared by the circuit breaker manufacturer for insertion into the books. The pages of each section shall be numbered consecutively. The sections shall be arranged in alphabetical order, but not necessarily lettered consecutively in the book.

3.29.3 Part 3 - miscellaneous switchboard devices. - Switchboard devices shall be arranged by sections as follows, prepared by the manufacturer. The pages of each section shall be numbered consecutively. The sections shall be arranged in alphabetical order, but not necessarily lettered consecutively in the chapter.

Chapter 1 - Instruments, instrument transformers and instrument accessories.
Section A - D.c. ammeters and voltmeters.
Section B - A.c. ammeters and voltmeters.
Section C - Wattmeter.
Section E - Frequency meter.
Section F - Synchroscope.
Section G[1] - Temperature meter equipment.
Section H - Instrument transformers.
Section J - Current transformer protective device.
Section K - Shunts.
Chapter 2 - Switches and devices. - A general write-up on control switches and indicator lights, test blocks and plugs, and bus disconnecting switches, giving descriptive data, illustrations, rating, without specific reference to any one particular switchboard.
Chapter 3 - Relays.
Section A - Protective relays.
Section B - Voltage control relays.
Section C - Current control relays.
Section D - Auxiliary relays.
Section E - Bus transfer (where not covered above).
Chapter 4 - Rheostats and miscellaneous.

[1]Separately mounted for switchgear groups not having a control benchboard.

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3.29.4 Part 4 - repair parts and tools.- The complete parts list may be a reduction of the list of repair parts conforming to Specification MIL-F-17292 or separately typed sheets.

3.29.4.1 Maintenance.- Maintenance of equipment shall be included in the section for each particular piece of equipment. These sections shall show complete information for enabling laymen to effect efficient maintenance of equipment, including service inspection, lubrication, and routine maintenance.

3.29.4.2 Drawings.- Drawings, illustrations, and photographs shall include the following:

(a) Representative photographs of switchboards, including front and rear view of each switchboard and typical open panel view.
(b) A front view arrangement plan of each switchboard showing in simplified fashion, arrangement, part identification and nameplate number of each device on front of switchboard. The nameplate identification shall agree with the switchboard plan. (See part 1, chapters 2 to 6 inclusive, herein.) This plan may be a reduction of the switchboard front view plan.
(c) A simplified one-line diagram showing only arrangement of primary connections for generators, bus ties, buses, typical feeders, and shore connections. Each circuit breaker or device shown on this diagram shall have nameplate identification related to (2) above. No control or metering circuits shall be shown except for very simple systems where the inclusion of control and metering circuits will not complicate the drawing. For simple systems this diagram may be a reduction of the switchboard complete wiring diagram.
(d) Metering and control diagram: This diagram may be reduction of the switchboard complete wiring diagram. When a specifically drawing diagram is furnished, it shall show the major bus circuits, the number and location of all instruments and control transformers, the instruments, relays and other equipment connected to each transformer, and the location of instruments in relation to the various switchboards.
(e) A diagram showing typical circuits, such as synchronizing, and ground lamps, to be separately illustrated and discussed in the text under part 1, chapter 1.

3.30 Repair parts.- Repair parts conforming to Specification MIL-R-15137 shall be supplied on a per vessel or per switchboard basis as specified (see 6.1) and shall include components for the following equipment in the quantities indicated:

(a) For types ACB and AQB - circuit breakers.- Repair parts shall conform to Specifications MIL-C-17587 and MIL-C-17361.
(b) For bus transfer equipment.- Repair parts shall conform to Specification MIL-S-17773.
(c) For tumbler switches.- One complete switch for each ten switches, or fraction thereof, of each frame size and type installed.
(d) For rotary type control and instrument switches.- One moving and one stationary contact, if renewable type is used, for each ten moving and stationary contacts, or fraction thereof installed, but not less than one complete set for the largest switch installed. One set of operating springs for each 10, or fraction thereof installed.
(e) For transformer protective devices.- One transformer protective device of each size and type for each ten, or fraction thereof installed.
(f) For relays - induction type.- One coil and one set of contacts and springs of each type and size for each ten, or fraction thereof installed (where it is simpler and more practical, a complete relay may be supplied).
(g) Auxiliary or plunger-type.- One coil and one set of contacts and springs of each type and size for each ten or fraction thereof installed (where it is simpler and more practical, a complete relay may be supplied).

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(h) For fuses. - One nonrenewable fuse for each five nonrenewable fuses, or fraction thereof of each type and size installed; except that not less than 15 fuses of any one type and size shall be required.

(i) For indicator lights. -

Disc, globes and lenses	10 percent of installed.
Lamps	200 percent of installed.
Lampholders	1 percent of installed.
Lights, indicators (where lampholder and transformer or resistor are integral)	1 complete indicator light for each 20 or fraction installed.
Transformers (mounted separately)	5 percent of installed.
Resistors (mounted separately)	5 percent of installed.

(j) One repair resistor of each type and size for each ten, or fraction thereof, installed.

(k) Dead front fuse holders. - One for each ten installed but not more than five.

In case of conflict between these requirements and any requirements of the material specifications, the above requirements shall apply.

3.30.1 An index list of repair parts conforming to Specification MIL-F-17292 shall be provided.

3.31 Tools. - Ordinarily no tools are required to be furnished with switchboards. Where the construction of the switchboards or any of the appurtenances installed thereon is such as to require the use of certain special tools for their proper service maintenance, one complete set of such special tools shall be furnished for each ship's service and emergency switchboard installed. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.)

3.32 The Government will furnish material as specified (see 6.1).

3.33 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which have been approved for listing on the applicable Qualified Products List on the date of invitation for bids or date of initiation of purchase action shall be utilized. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 General. - The equipment and the preparation for delivery thereof shall be given such tests and inspection as will verify and assure conformance with this specification. All materials and components, both electrical and structural, shall be examined and tested in accordance with the applicable specifications either at the various sources or at the place of assembly.

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4.3 Inspection at the place of manufacture.-

4.3.1 Examination and tests of assembly.- Switchboards shall be completely assembled by the manufacturer in his works, and shall then be examined to assure compliance with this specification and tested to discover any incorrect connections, open circuits, short circuits, grounded circuits, and incorrect operation of the various apparatus. Dielectric strength test shall be made and the resistance of insulation shall be measured. The Government inspector shall witness the tests and perform visual and dimensional examination on all switchboards and panels selected in accordance with table VI, to verify conformance to this specification.

Table VI - Sampling for examination.

Lot size, number of switchboards or panels	Sample	Acceptance number (defectives)	Rejection number (defectives)
40 and under	7	0	1
41 to 65	10	0	1
66 to 110	15	1	2
111 to 180	25	1	2
181 to 300	35	2	3

4.3.1.1 Any board or panel having one or more defects or failing in any test shall be rejected, and if the number of such boards in any sample exceeds the acceptance number for that sample, the entire lot shall be rejected. Rejected lots may be offered again after the manufacturer has removed all nonconforming units. Samples shall again be selected and inspected in accordance with table VI, for the defect or failure which caused rejection, or for other defects also.

4.3.2 Structural rigidity.- The natural frequencies of vibration specified in 3.12.10 are those frequencies at the top of the framework as measured in the two directions in the plane of the top. These vibrations may be excited in several methods; however, for purpose of definition, they are those vibrations that result when the base of the unit is bolted to a rigid foundation and the top of the unit is deflected in a direction in its plane and suddenly released. Other methods of exciting this type of vibration may be by an oscillator of the form of an adjustable speed motor driving an unbalanced rotating mass, or simply by directing a low velocity blow at the top of the unit by use of a mass of sufficient size to cause 0.025 or 0.030 inch deflection. In all cases the resulting frequencies of vibration can be determined by any reliable vibration meter. The number of units to be tested by the manufacturer in the manner specified herein shall be one complete ship's service switchgear group and one emergency switchgear group for one ship of each class.

4.3.3 Dielectric strenth.- The equipment shall withstand for a period of 1 minute a dielectric test voltage of twice the rated voltage of the equipment plus 1,000 volts r.m.s. value. The frequency of the test voltage shall be not less than the rated frequency of the equipment and in no case less than 60 cycles. It shall approximate a true sine wave. The source of the test potential shall have a rating of at least 1 kw.

4.4 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Switchboards shall be shipped with instruments mounted on the panels. Switchboards and repair parts shall be prepared for the level of shipment, as specified (see 6.1), in accordance with Specification MIL-P-16018.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type (see 1.2), number, and arrangement of units comprising the switchgear sections (see 3.3 to 3.11, inclusive).
(c) Sections comprising a switchgear group (see 3.3. to 3.11, inclusive).
(d) Power, current, and voltage rating of generators controlled (see 3.3.1, 3.3.2, 3.3.9 and 3.8).
(e) Feeders, breakers, and/or switches (see 3.3.6, 3.3.7, 3.3.8, 3.5, 3.6, 3.7.3 and 3.7.4).
(f) Requirement for type AQB circuit breaker on emergency switchgear for Diesel cooling water booster pump (see 3.4.1).
(g) Requirement for electrically operated emergency bus unloader switch (see 3.4.1 and 3.4.2).
(h) Rating for shore power connection circuit breaker (see 3.3.3 (h), 3.3.4.2 (a) and 3.7.2.2).
(i) Limiting overall dimensions (see 3.12.2).
(j) Material to be used in construction of switchgear structure and switchgear buses (see 3.12.11 and 3.17).
(k) Busses to be provided in each switchgear section (see 3.17).
(l) Location of cable entrance (see 3.17.18).
(m) Circuit breaker frame size (see 3.19).
(n) Circuit breaker trip element ratings and tripping characteristics (see 3.19).
(o) The following items are not a part of any individual standard unit. They will be supplied by the switchboard manufacturer only when complete switchgear sections are ordered. Otherwise they will be supplied by the agency assembling the standard units into switchgear sections:
    (1) Foundation channels.
    (2) Top sheet and drip shield.
    (3) Main bus bars.
    (4) Main bus supports, main bus spreaders and bus opening cover on end units.
    (5) Cable and wiring between units.
    (6) End trims.
(p) List of Government furnished material (see 3.62).
(q) Bases for repair parts (per vessel or per switchboard) (see 3.30).
(r) Level of shipment required (see 6.1).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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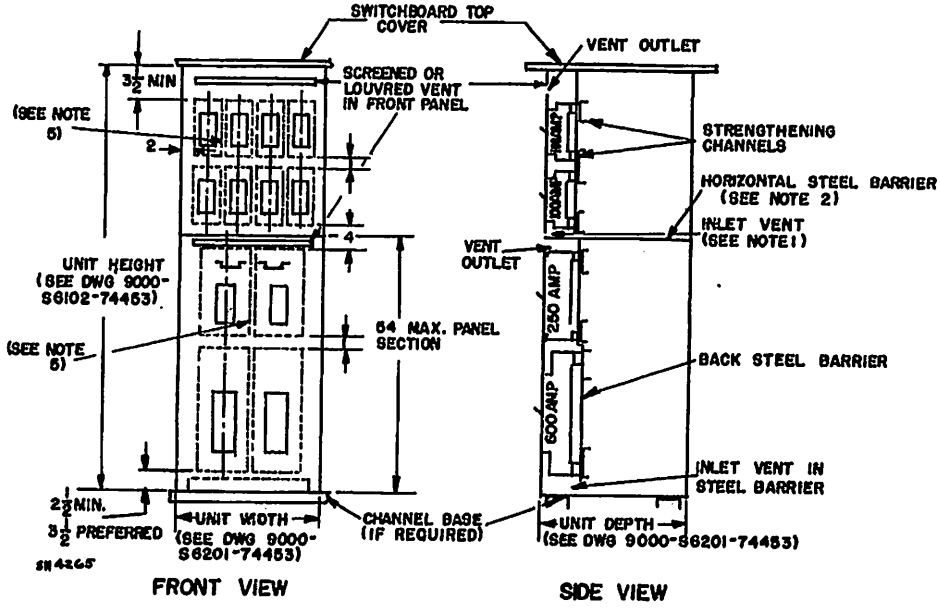


Figure 1 - Distribution unit type AQB circuit breakers (dimensions and spacings).

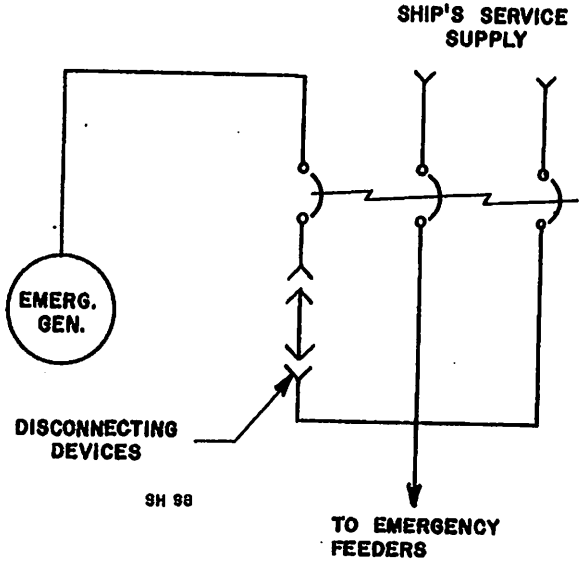


Figure 2 - Three-way bus transfer type ABT-A3(ES), circuit breaker type.

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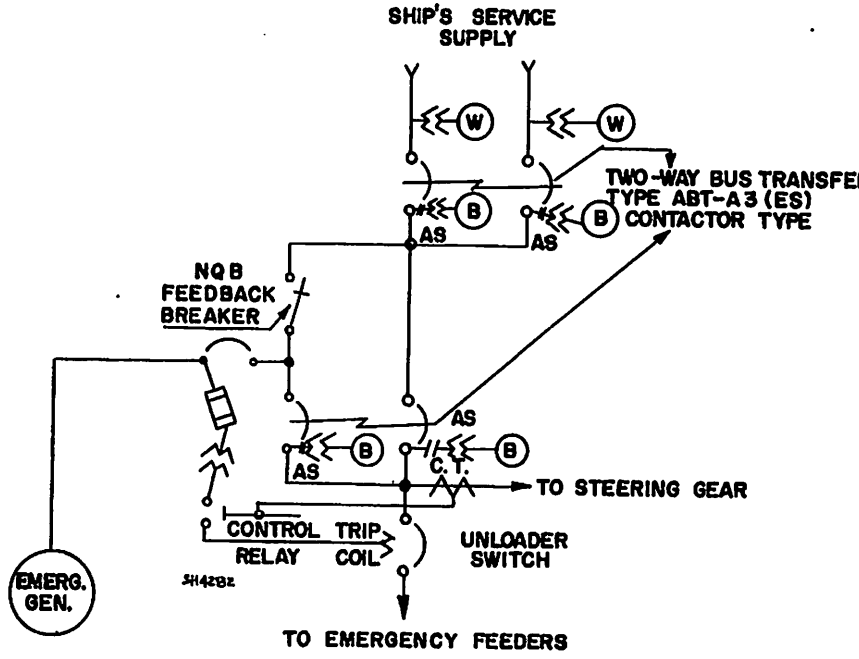


Figure 3 - One line diagram of emergency power system.

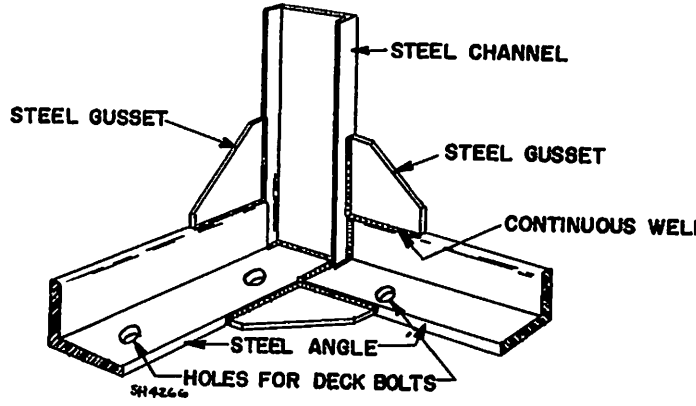
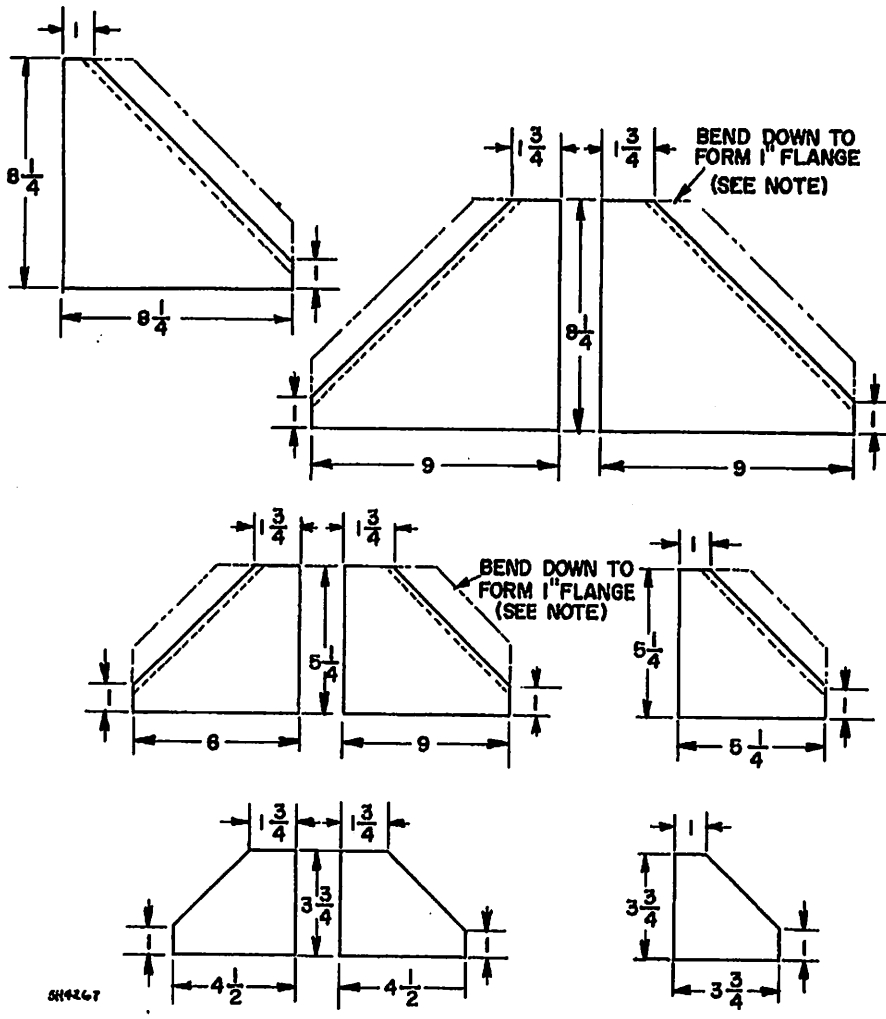


Figure 4 - Typical bottom corner construction.

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NOTE: Flanges shall be used when thickness of gusset plates is less than 3/16 inch. Gusset plates shall not be less than 1/8 inch thick.

Figure 5 - Typical gusset plates.

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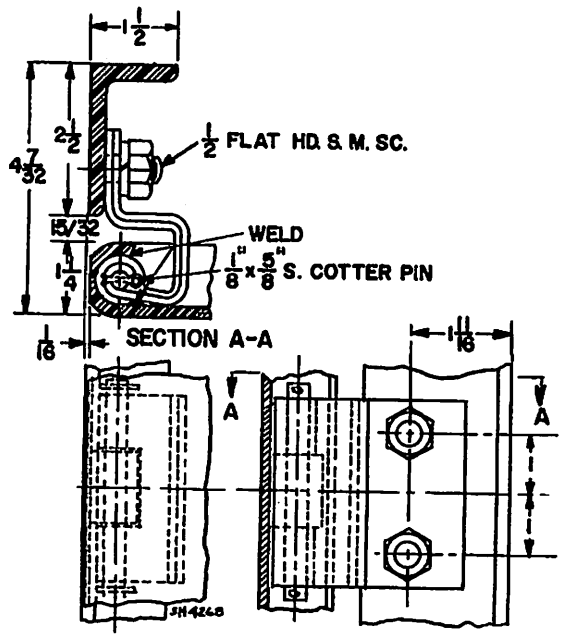


Figure 6 - Typical panel hinge assembly.

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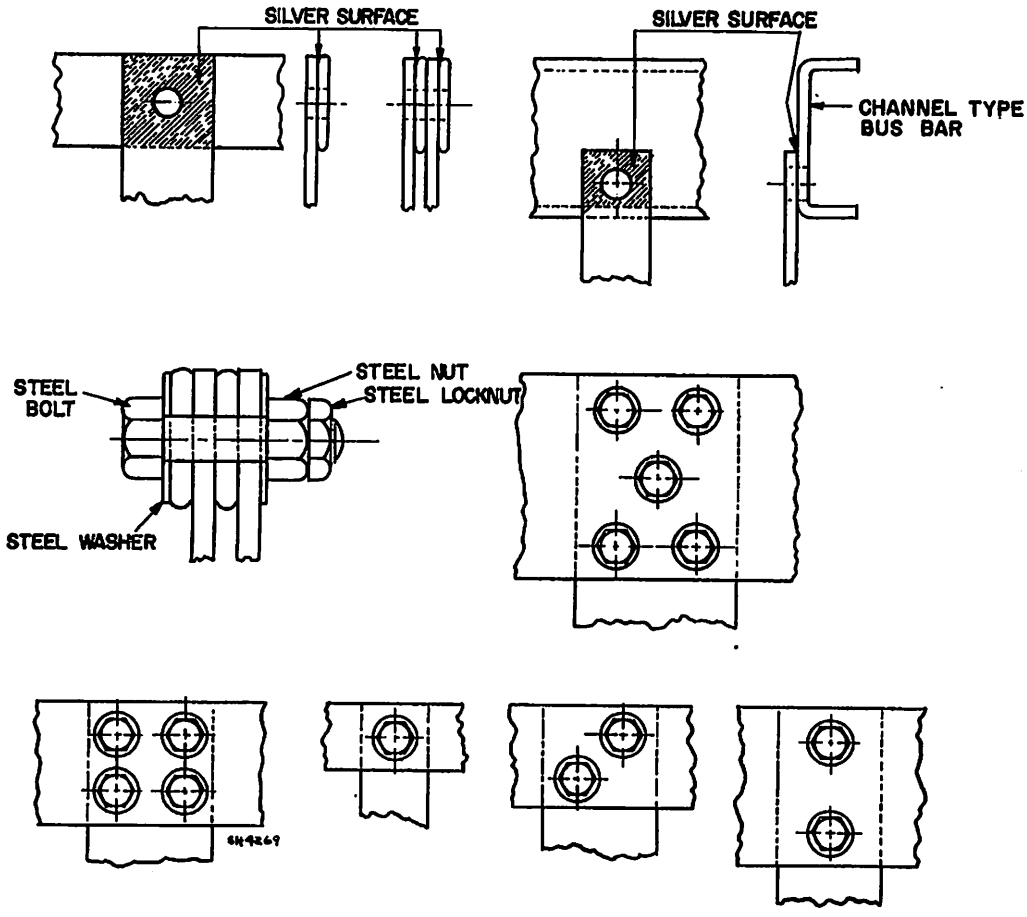


Figure 7 - Arrangement for bolting of bus connections. (For combination of bolt and bus sizes see table IV.)

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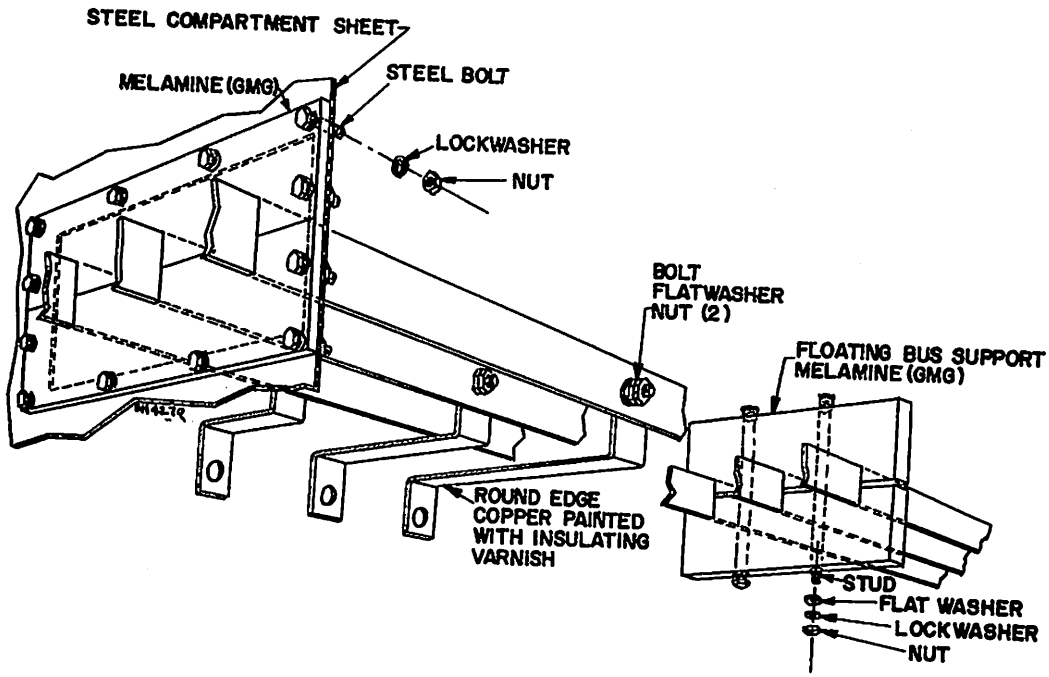
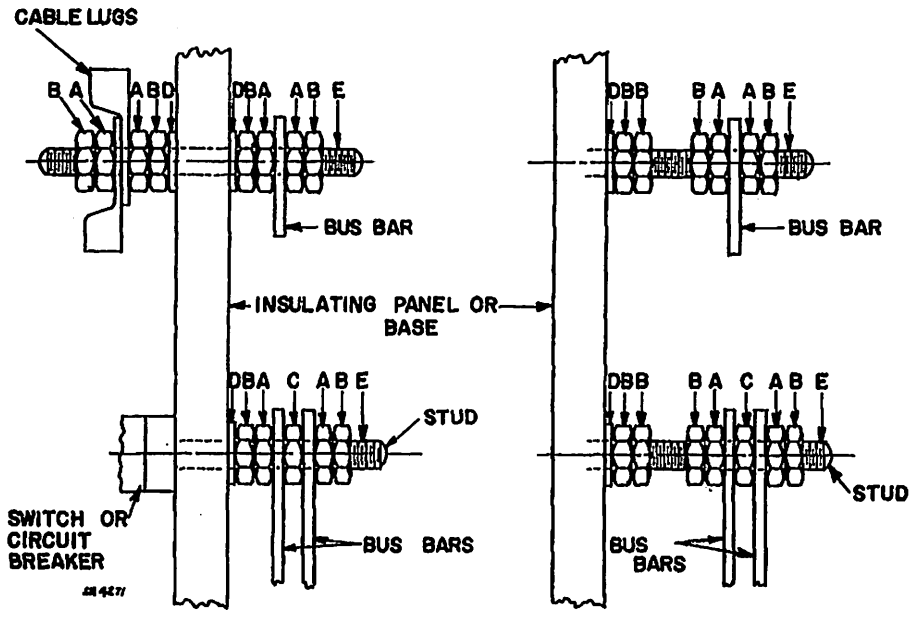


Figure 8 - Typical bus bar mountings.

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A. – ALL CONTACT NUTS SHALL BE BRASS, SILVER SURFACED. (TO BE OF SUFFICIENT SIZE TO PROVIDE NECESSARY CONTACT AREA.)
B. – ALL LOCKNUTS TO BE SAME AS CONTACT NUTS.
C. – ALL CONTACT SPACER NUTS TO BE SAME AS CONTACT NUTS.
D. – BRASS WASHER.
E. – ALL STUDS TO BE COPPER AND SHALL BE SILVER SURFACED.

Figure 9 - Stud contacts with copper and aluminum buses.

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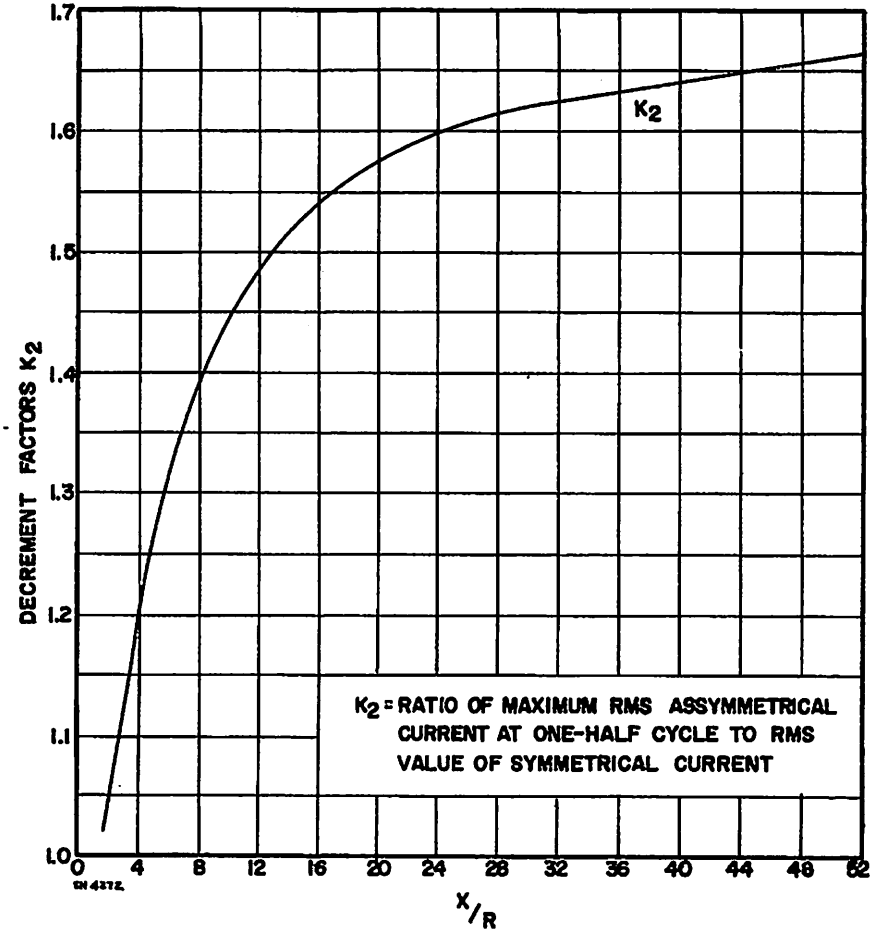
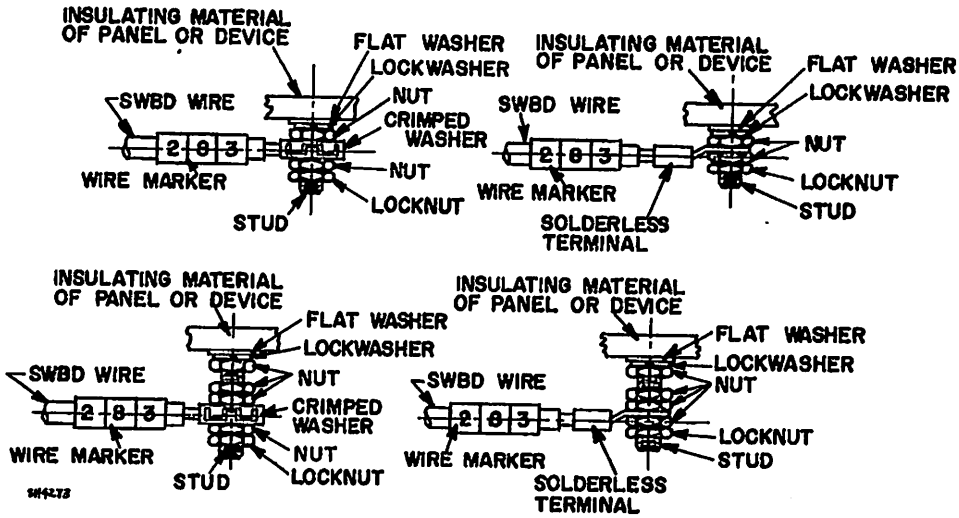


Figure 10 - Factors for converting r.m.s. value of symmetrical current to maximum r.m.s. asymmetrical in three phases.

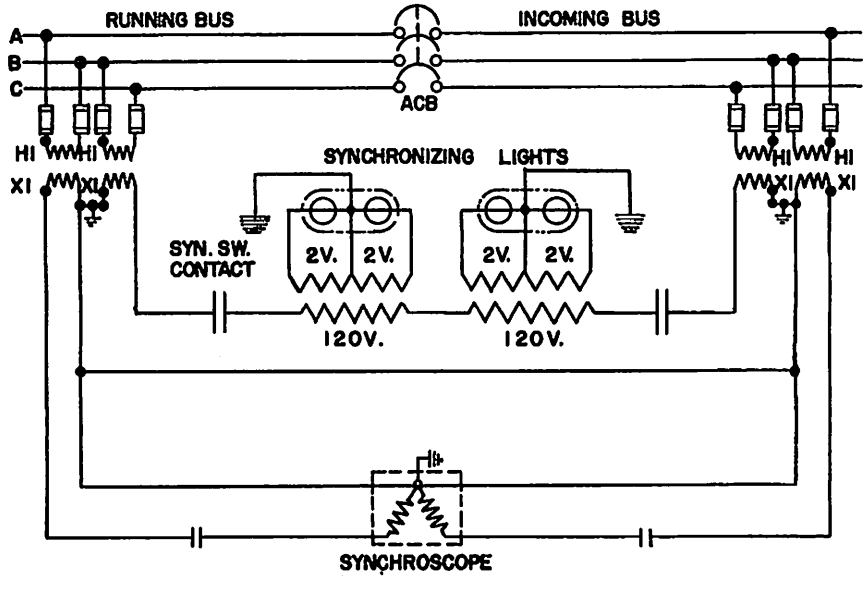
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:16Z
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NOTE: MARKINGS MAY BE STAMPED ON WIRE TERMINALS OR, IF TERMINALS ARE TOO SMALL TO PERMIT STAMPING, SYNTHETIC RESIN TUBING, SPECIFICATION MIL-I-631, TYPE F, GRADE A, OR FIBER TAGS SHOULD BE USED.

Figure 11 - Cable marking and detail of connections.

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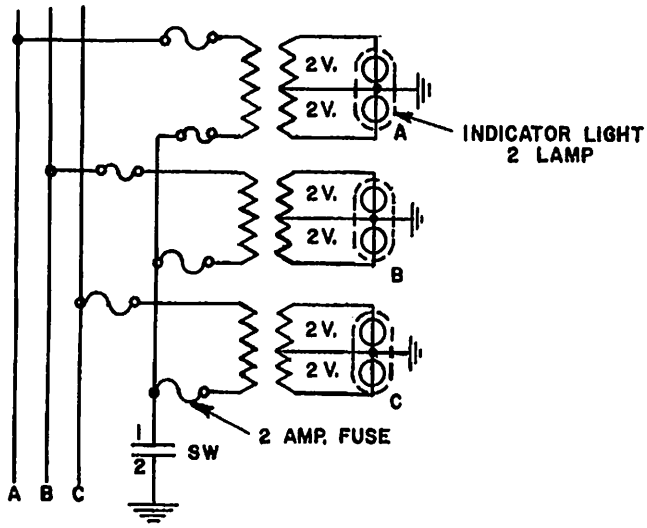


NOTE: THIS DIAGRAM IS TYPICAL ONLY, REPRESENTING A REFERENCE BUS AND A SECOND BUS TO BE SYNCHRONIZED WITH IT. WHERE PROVISION MUST BE MADE FOR A THIRD BUS TO BE SYNCHRONIZED WITH EITHER OF THESE BUSES, ADDITIONAL CONTACTS IN THE SYNCHRONIZING TRANSFER SWITCH SHALL BE PROVIDED AS REQUIRED.

SH 99

Figure 12 - Synchronizing connections.

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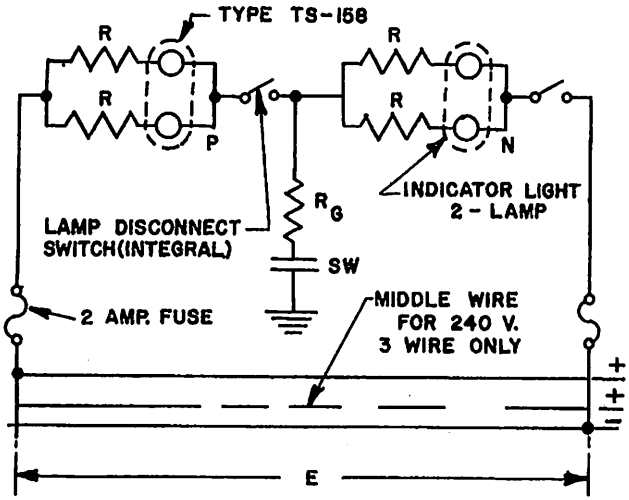
A.C. GROUND LIGHTS
450 V. – 3 PHASE

NORMAL – ALL LIGHTS DIM
PHASE A GRD – "A" LIGHT DARK
"B" LIGHT BRIGHT
"C" LIGHT BRIGHT
PHASE B GRD – "B" LIGHT DARK
"A" LIGHT BRIGHT
"C" LIGHT BRIGHT
PHASE C GRD – "C" LIGHT DARK
"A" LIGHT BRIGHT
"B" LIGHT BRIGHT

CONTACTS		POSITIONS	
		TEST	NORMAL
	1	X	
	2	X	

SPRING RETURN

A.C. GRD. LIGHTS
450 & 117V.-3 PHASE



E (VOLTS)	R (OHMS)	R_G (OHMS)
120	1100	630
240	2800	1400

SH 100

D.C. GROUND LIGHTS
120 V. – 2 WIRE AND
120/240 V. – 3 WIRE

NORMAL – BOTH LIGHTS DIM
POS. GRD. – "P" LIGHT DARK
"N" LIGHT BRIGHT
NEG. GRD. – "N" LIGHT DARK
"P" LIGHT BRIGHT

CONTACTS		POSITIONS	
		TEST	NORMAL
	1	X	
	2	X	

SPRING RETURN

D.C. GRD. LIGHTS
120V. – 2 WIRE

NORMAL – BOTH LIGHTS DIM
POS. GRD. – "P" LIGHT DARK
"N" LIGHT BRIGHT
NU. GRD. – "P" LIGHT DIM
"N" LIGHT DARK
NEG. GRD. – "P" LIGHT BRIGHT
"N" LIGHT DARK

CONTACTS		POSITIONS		
		NEU.	NORMAL	POS-NEG
	1		X	X
	2	X		X

SPRING RETURN

D.C. GRD. LIGHTS
120/240V. – 3 WIRE

Figure 13 - Ground light schemes.

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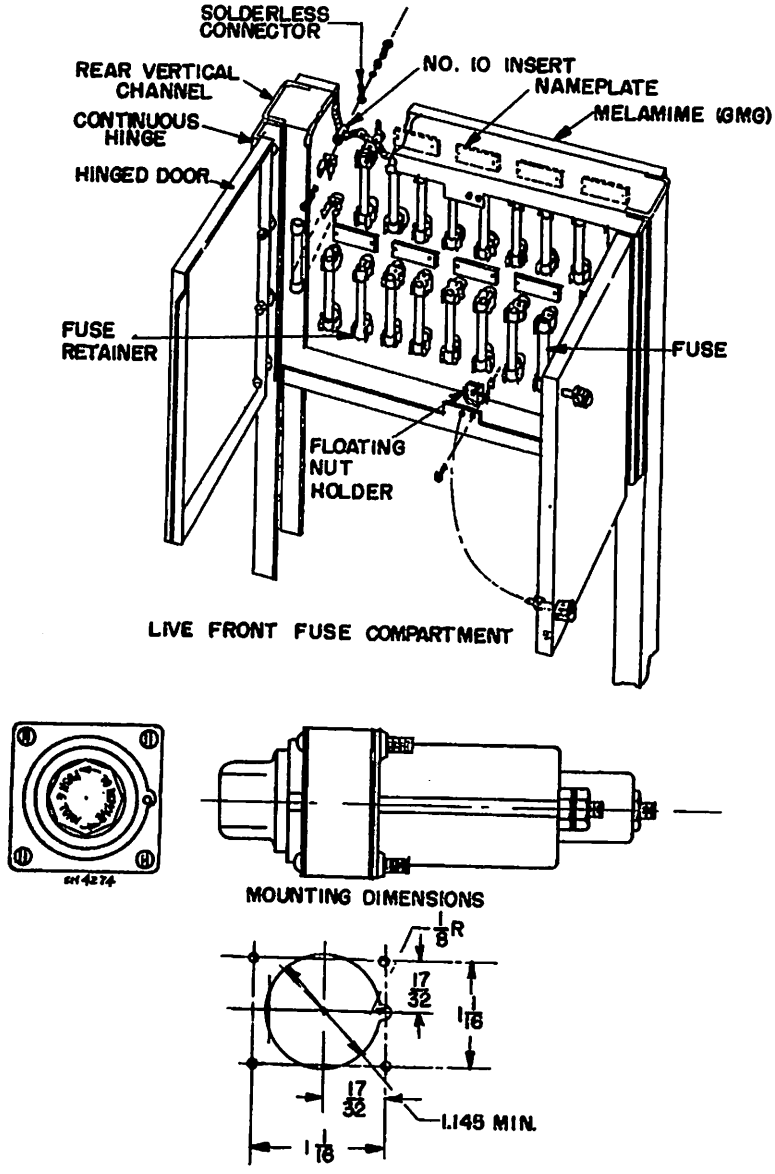


Figure 14 - Fuse mounting.

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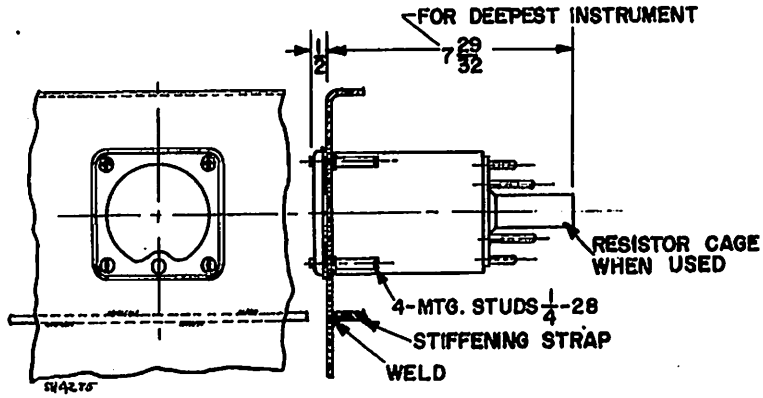


Figure 15 - Method of mounting switchboard instruments.

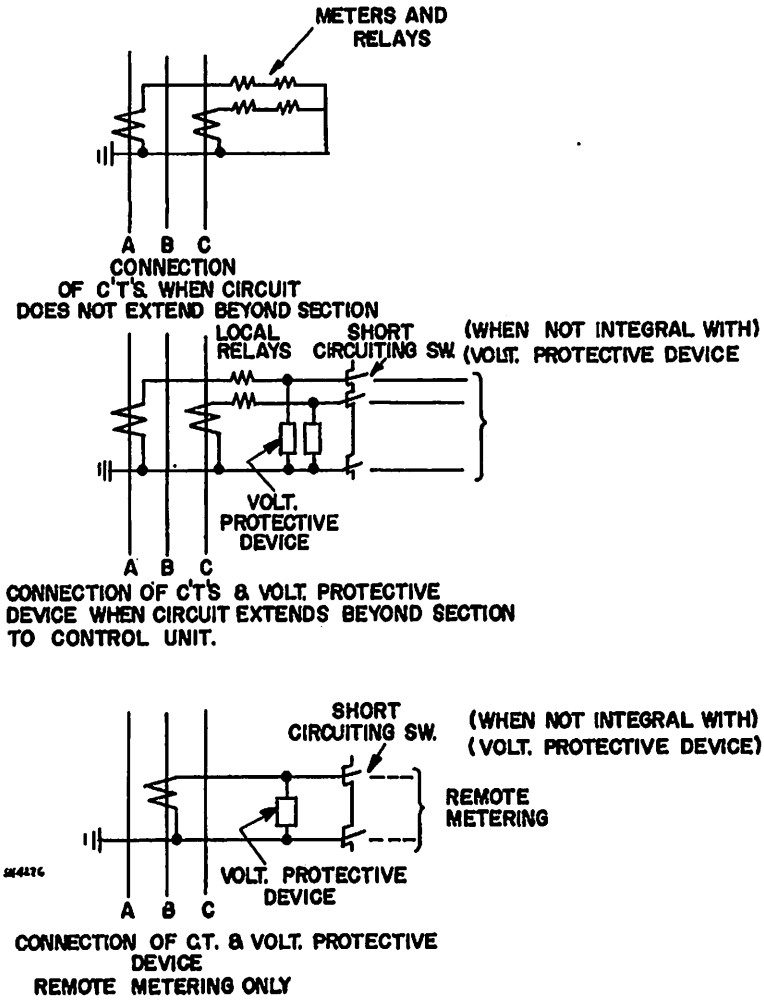


Figure 16 - Connections of current transformers and voltage protective devices.

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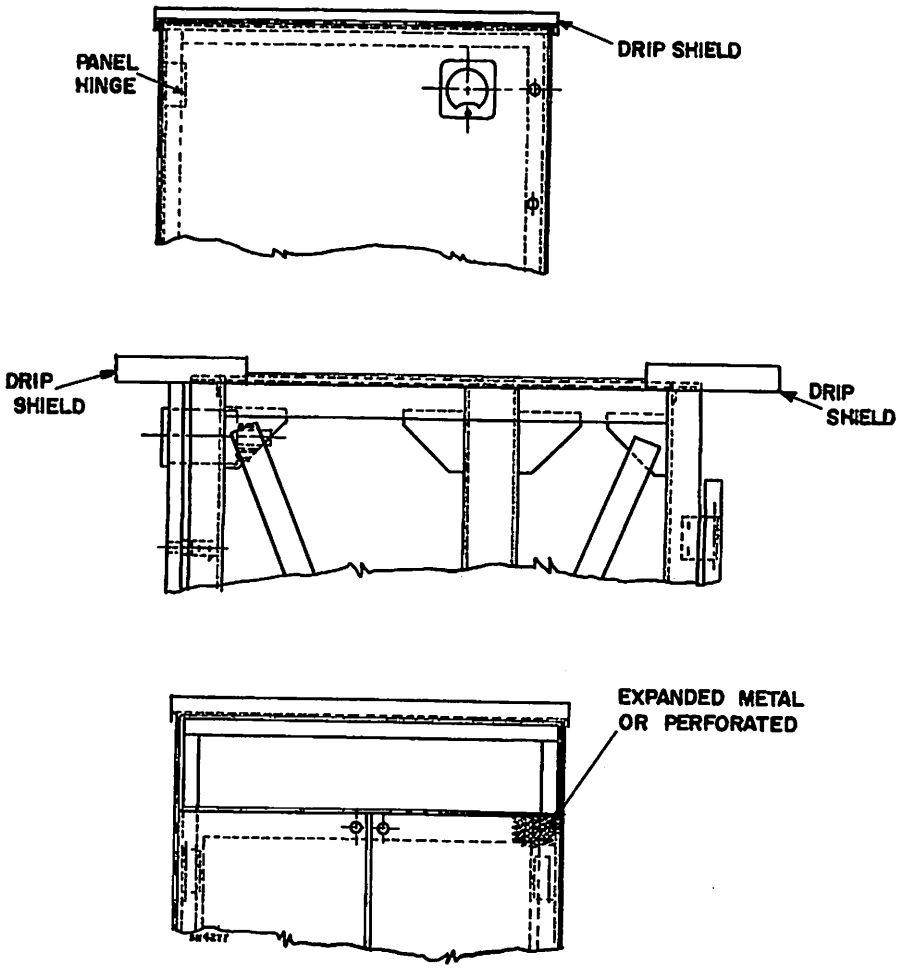


Figure 17 - Typical drip shield.

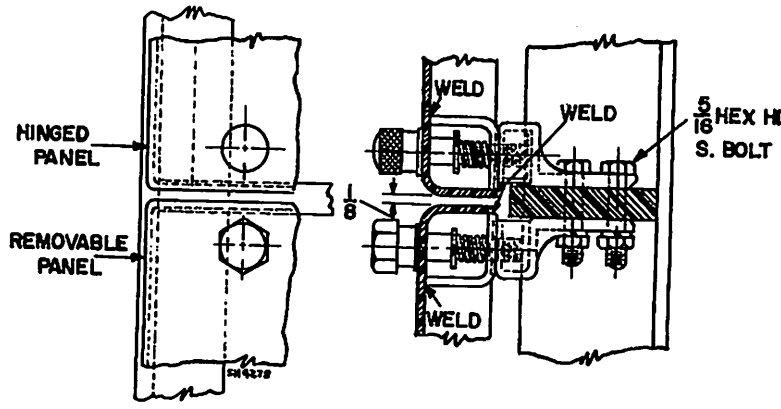


Figure 18 - Panel latch bolt for hinged and removable panel.

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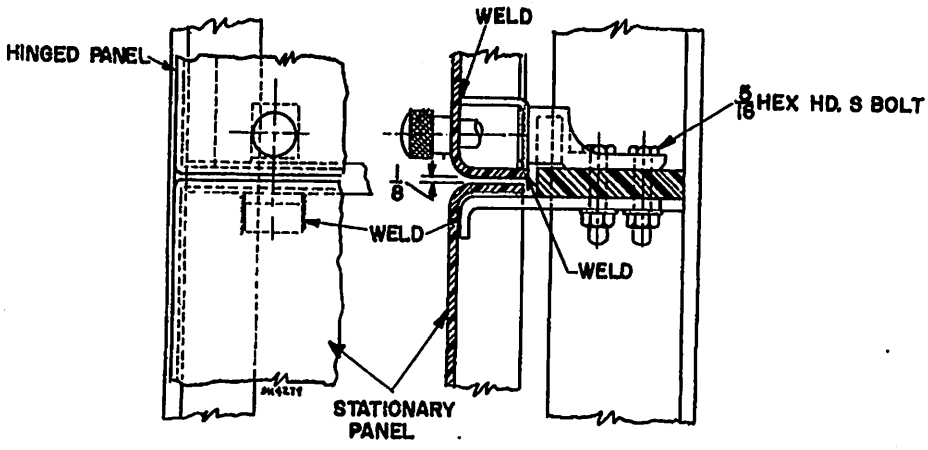


Figure 19 - Panel latch bolt for hinged panel and method of securing stationary panel.

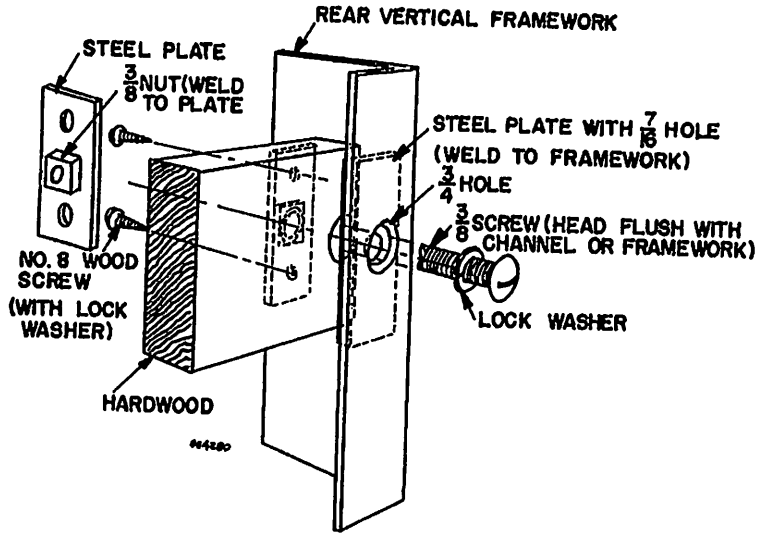


Figure 20 - Typical rear guard rail mounting.

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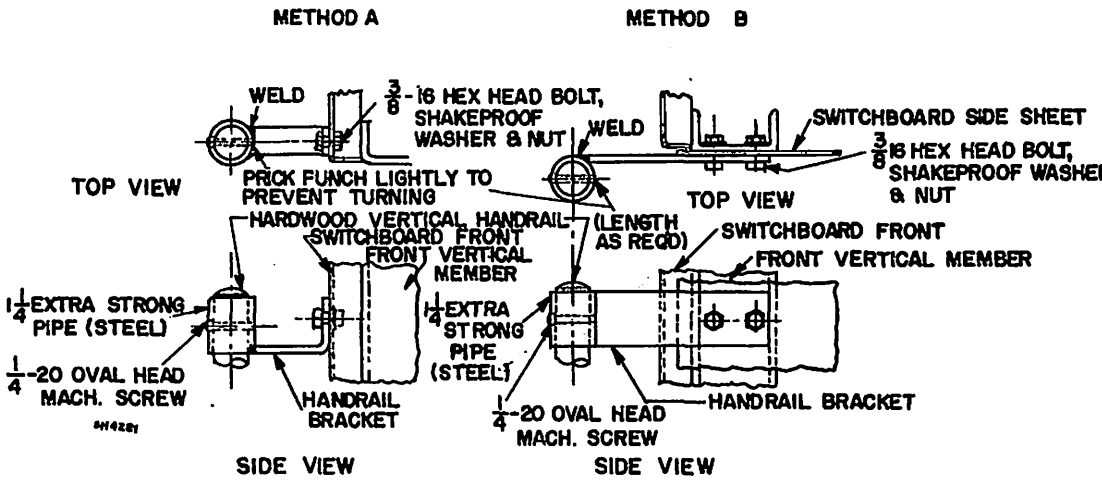


Figure 21 - Typical vertical guard rail mounting.

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